



Embedded AI-Driven Soft Multimodal Single-Electrode Sensor for Proximity, Tactile, and 2D-Strain Sensing

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Korea University

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- 1.2. Soft Inductive Sensor
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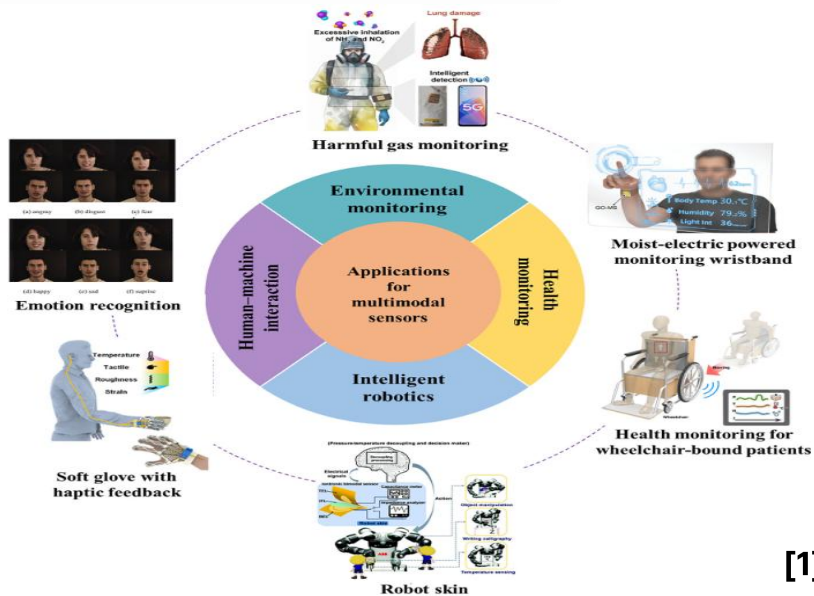
- 2.1. Setup & Preparation

3. Discussion

- 3.1. Signal Decoupling Analysis
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Introduction – Soft Multimodal Sensor

Need for Soft Multimodal Sensing



[1]

1. Soft Sensors

- Advances in robotics and human-machine interaction require human-like perception.
- Flexible sensors enable conformal contact, large deformation, and wearable applications.

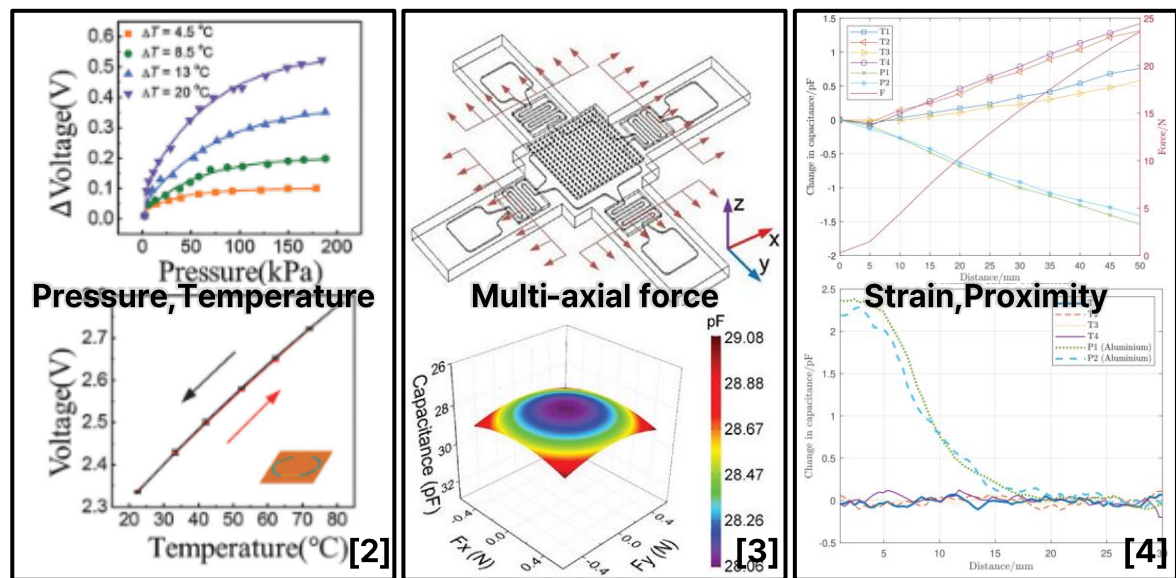
2. Multimodal Sensing

- Unimodal sensors detect only one physical parameter, limiting real-world usability.
- Real interactions involve multiple stimuli simultaneously.

Fundamental Challenge

Signal Coupling

Different physical stimuli generate overlapping electrical responses.



- [2] Pressure vs. Temperature → Resistance drift
[3] Multi-axis forces → Overlapping signal components
[4] Strain vs. Proximity → Capacitance change

[1] "Recent Progress on Flexible Multimodal Sensors: Decoupling Strategies, Fabrication and Applications." Advanced Materials (2026)

[2] "Electronic Skin with Multifunction Sensors Based on Thermosensation." Advanced Materials (2017)

[3] "Highly sensitive linear triaxial force sensor based on multimodal sensing for 3D pose reconstruction." Small Methods (2025)

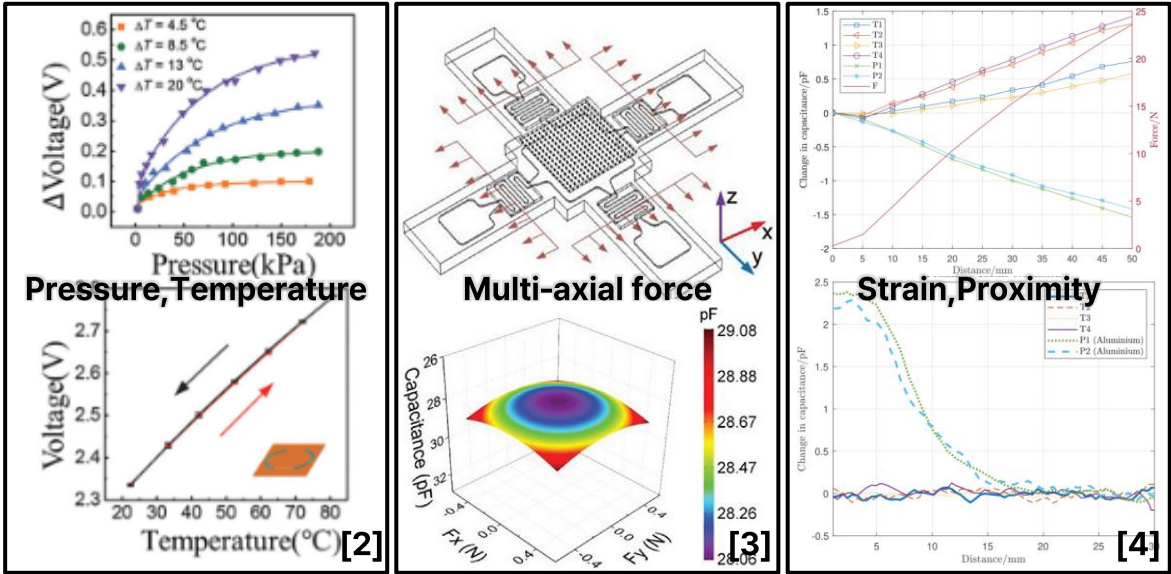
[4] "Highly stretchable additively manufactured capacitive proximity and tactile sensors for soft robotic systems." IEEE Transactions on Instrumentation and Measurement (2023)

Introduction – Soft Multimodal Sensor

Fundamental Challenge

Signal Coupling

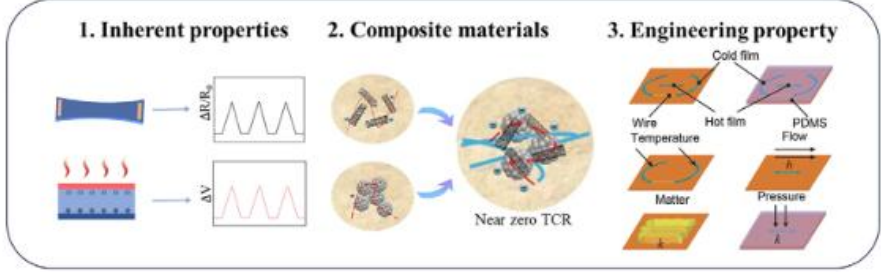
Different physical stimuli generate overlapping electrical responses.



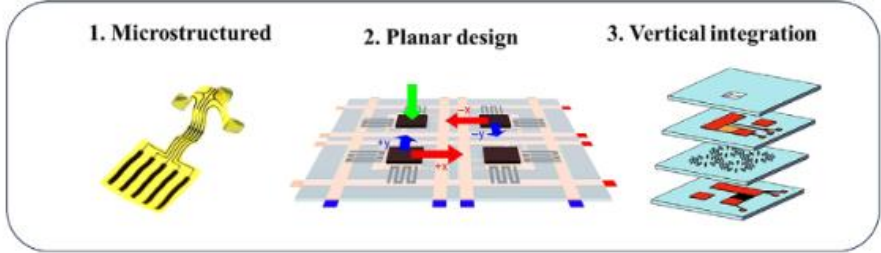
- [2] Pressure vs. Temperature $\rightarrow$ Resistance drift
- [3] Multi-axis forces $\rightarrow$ Overlapping signal components
- [4] Strain vs. Proximity $\rightarrow$ Capacitance change

Conventional Decoupling Strategies

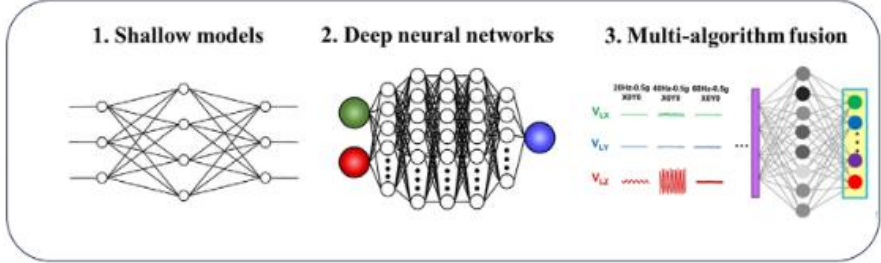
1. Material Design Decoupling



2. Structural Design Decoupling



3. AI-Based Signal Decoupling



[5]

[5] "Multimodal sensors with decoupled sensing mechanisms." Advanced Science 9.26 (2022)

Introduction – Soft Multimodal Sensor

Conventional Decoupling Strategies

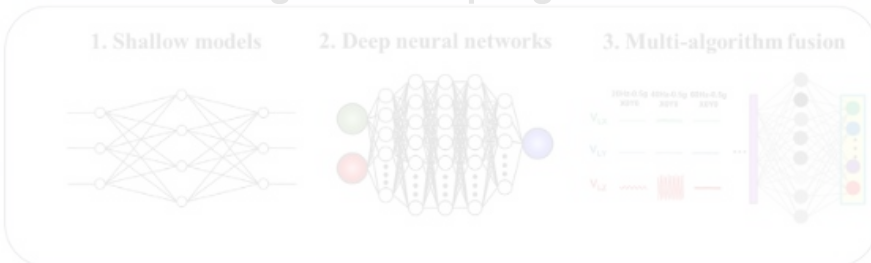
1. Material Design Decoupling



2. Structural Design Decoupling



3. AI-Based Signal Decoupling



[5]

Limitations

- **Complex** synthesis and optimization
- Only separates **intrinsically independent** signals
- Requires **bulky** and **expensive** measurement systems

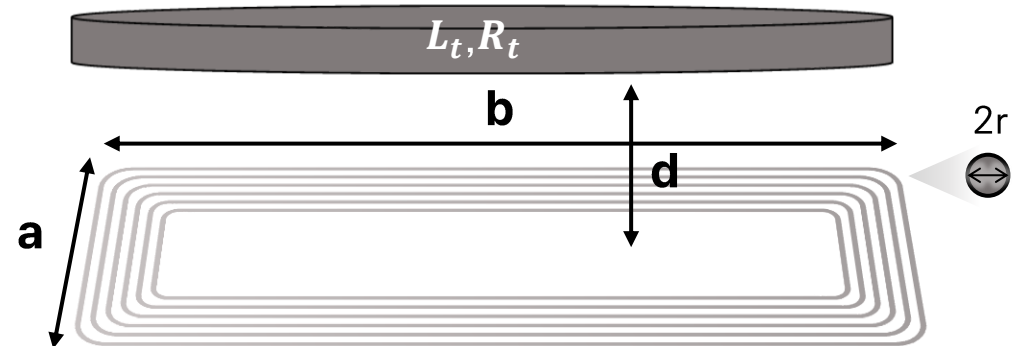
- **Complex** structural fabrication
- Increased **thickness** and **delamination risk** between integrated sensors
- Excessive **wiring** complexity
- Limited to **independent signal** separation

- Demands **complex** multi-sensor arrays
- Requires **bulky** and **expensive** measurement systems
- Requires **massive** labeled **datasets** for training
- **High** computational **cost** (poor real-time edge deployment)

Embedded AI-Driven Soft Multimodal Single-Electrode Sensor

Introduction – Soft Inductive Sensor

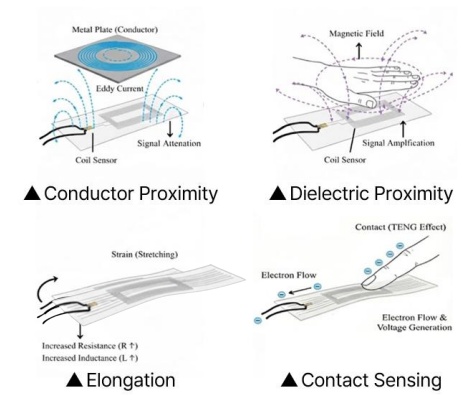
Electromagnetic Characteristics



$L(\epsilon, d)$: Function of **strain** & **target distance**
 $R_S(\epsilon, d)$: Function of **strain** & **target distance**
 $R_{DC}(\epsilon)$: Function of **strain**

* L : Inductance
* $R_S(\epsilon, d)$: AC Equivalent Series Resistance
* $R_{DC}(\epsilon)$: DC Resistance

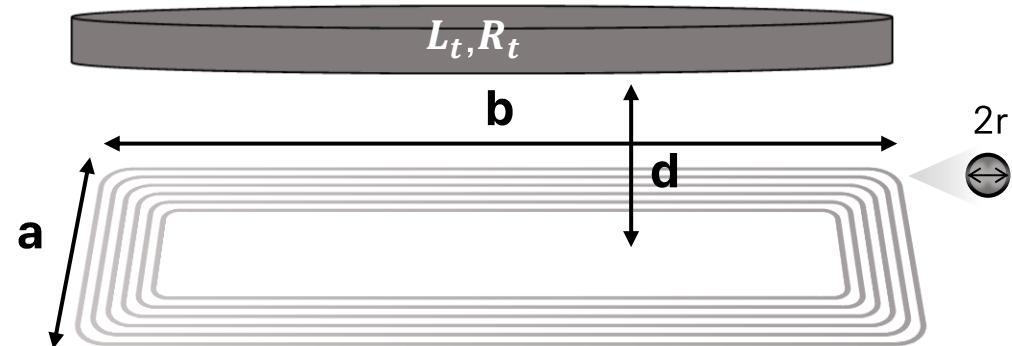
	L	R_S	R_{DC}
Stretching	↑	↑	↑
Conductor Approach	↓	↑	–
Dielectric (Skin) Approach	↑	–	–
Charged Object Approach	V ↑		



[1] "Programming stretchable planar coils as strain-invariant inductors and ultrasensitive wearable sensors." Materials Today Physics (2025)
[2] "Calculation and simulation of impedance diagrams of planar rectangular spiral coils for eddy current testing." NDT & e International (2006)

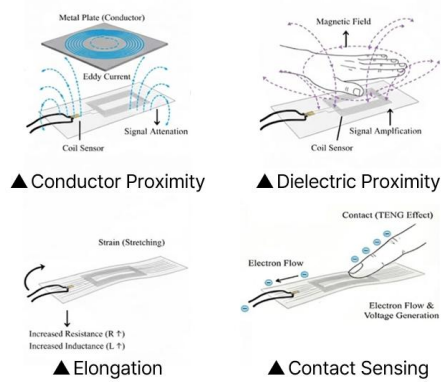
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Elongation

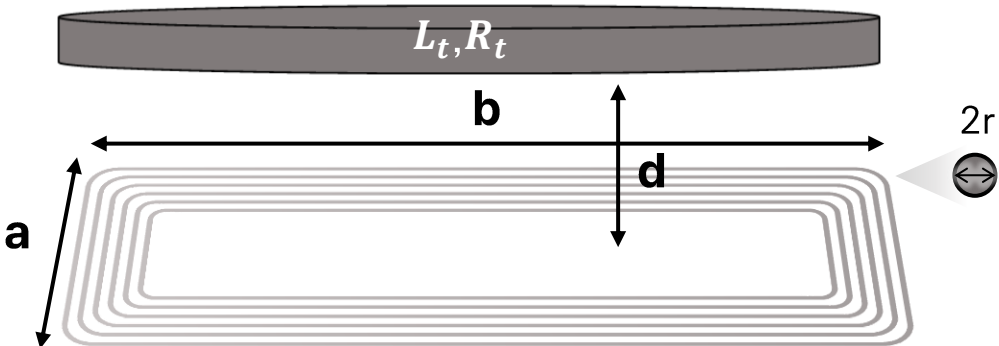
$L, R_{DC}, R_S \uparrow$



[1] "Programming stretchable planar coils as strain-invariant inductors and ultrasensitive wearable sensors." Materials Today Physics (2025)
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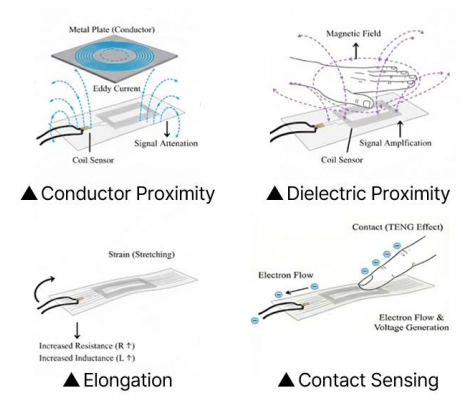
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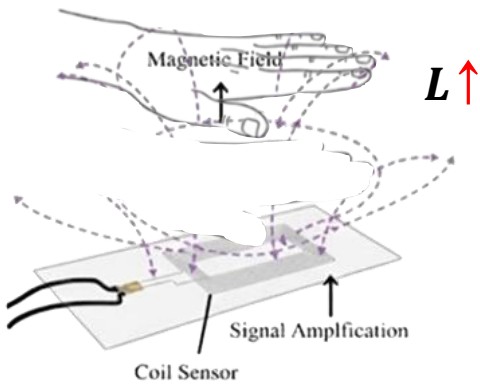
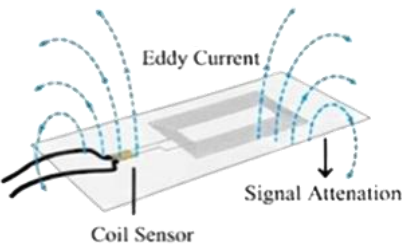
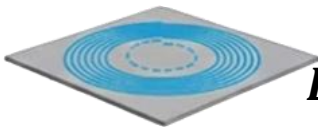
Elongation

$L, R_{DC}, R_S \uparrow$



Object Approach

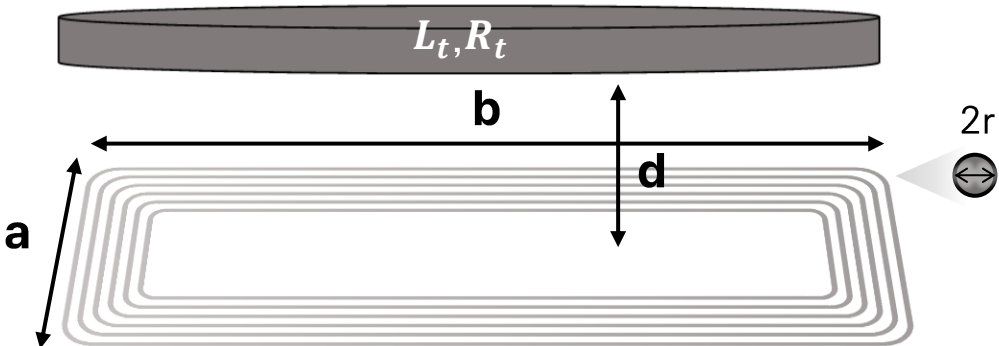
$L \downarrow, R_S \uparrow$



[1] "Programming stretchable planar coils as strain-invariant inductors and ultrasensitive wearable sensors." Materials Today Physics (2025)
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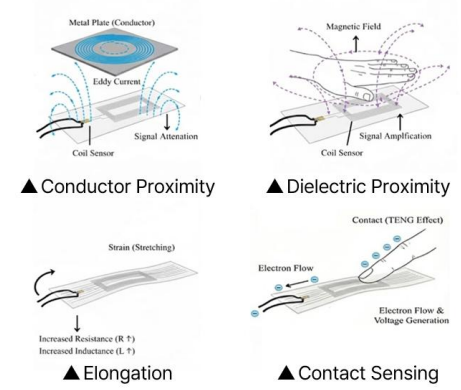
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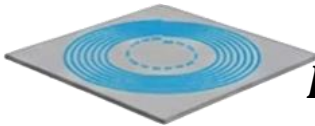


Elongation

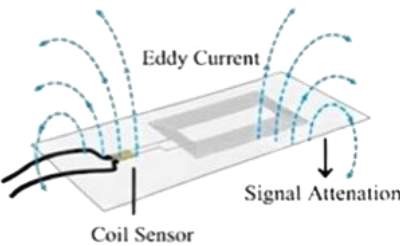
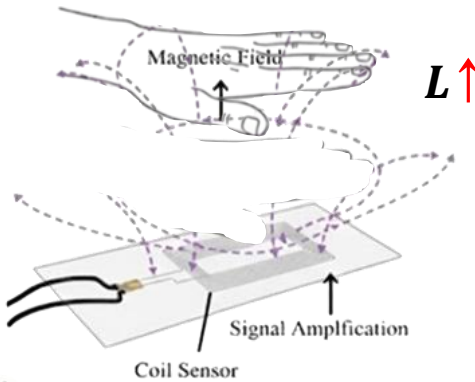
$L, R_{DC}, R_S \uparrow$



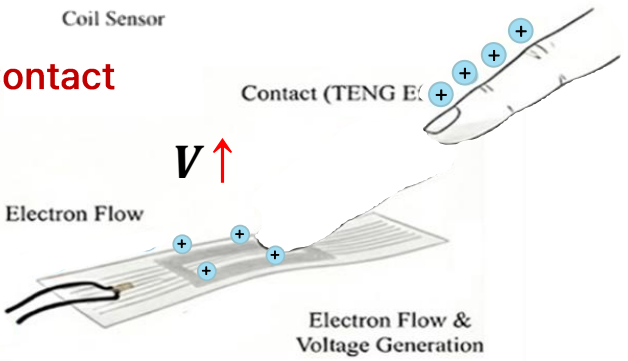
Object Approach



$L \downarrow, R_S \uparrow$



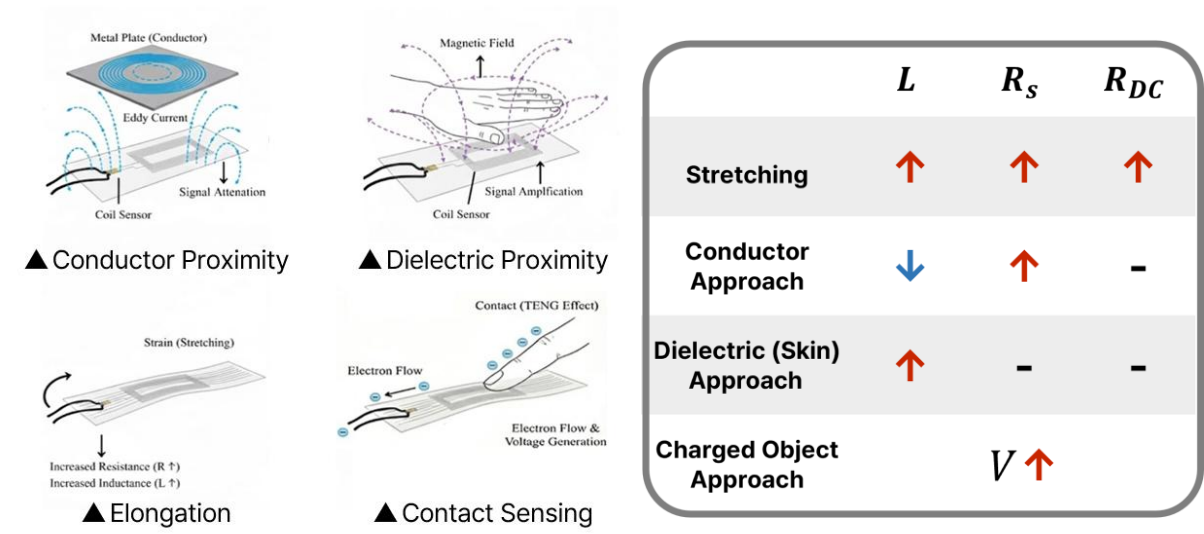
Contact



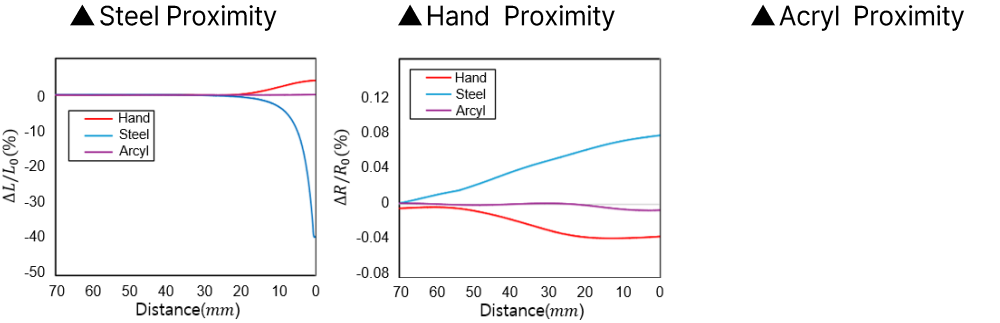
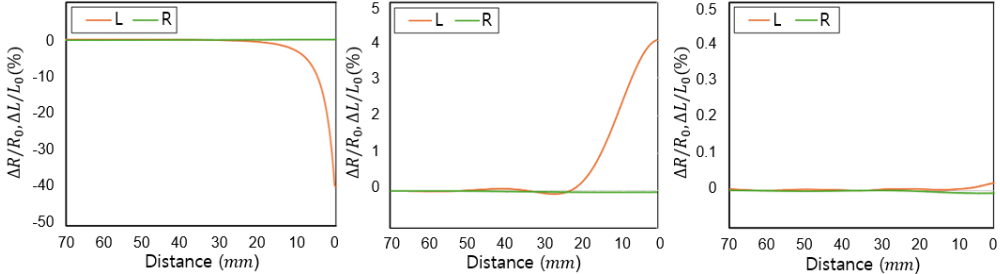
[1] "Programming stretchable planar coils as strain-invariant inductors and ultrasensitive wearable sensors." Materials Today Physics (2025)
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Introduction – Soft Inductive Sensor

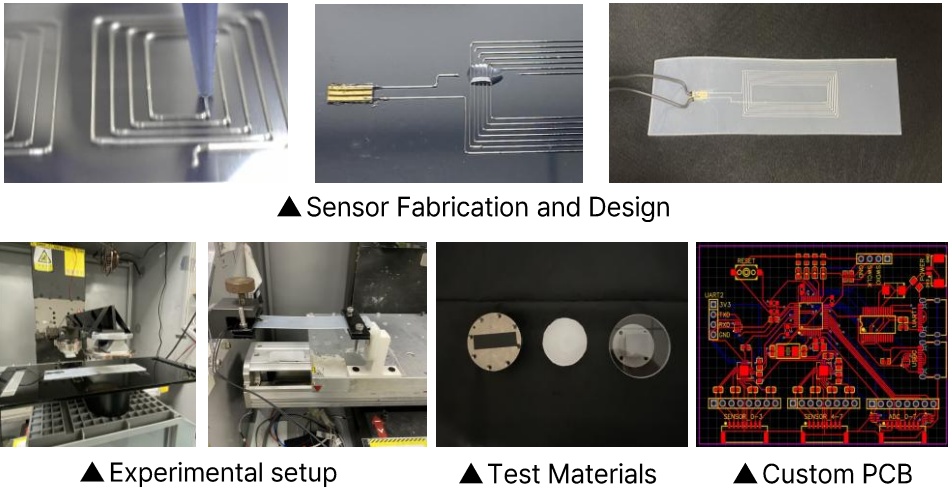
Electromagnetic Characteristics



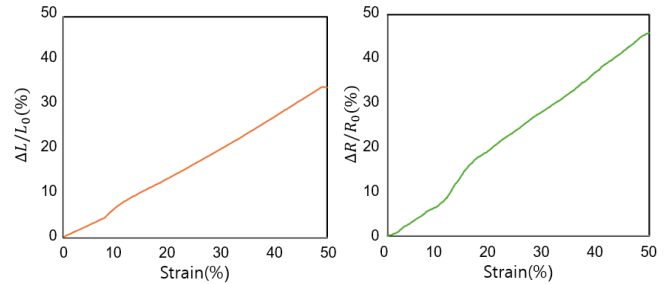
< Proximity >



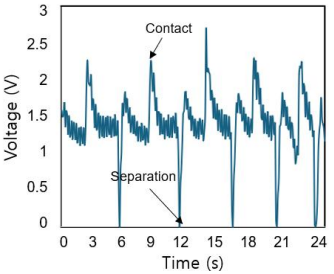
Sensor Response Test Setup



< Strain >

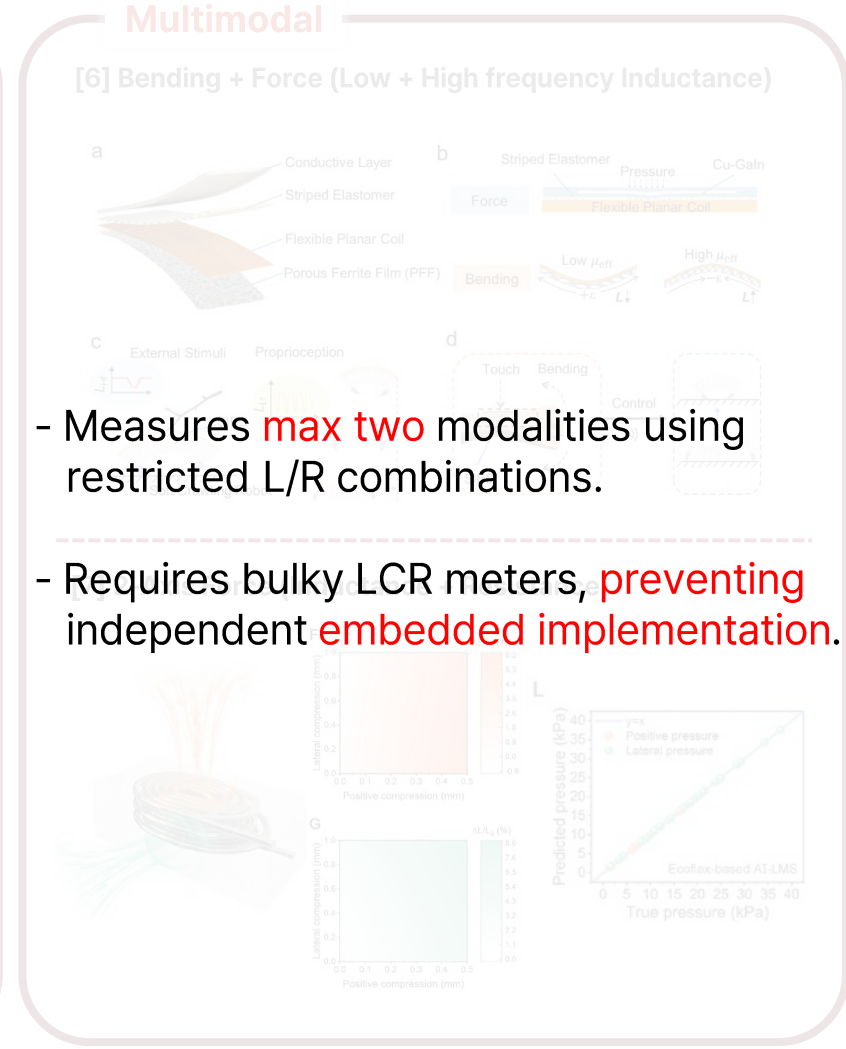
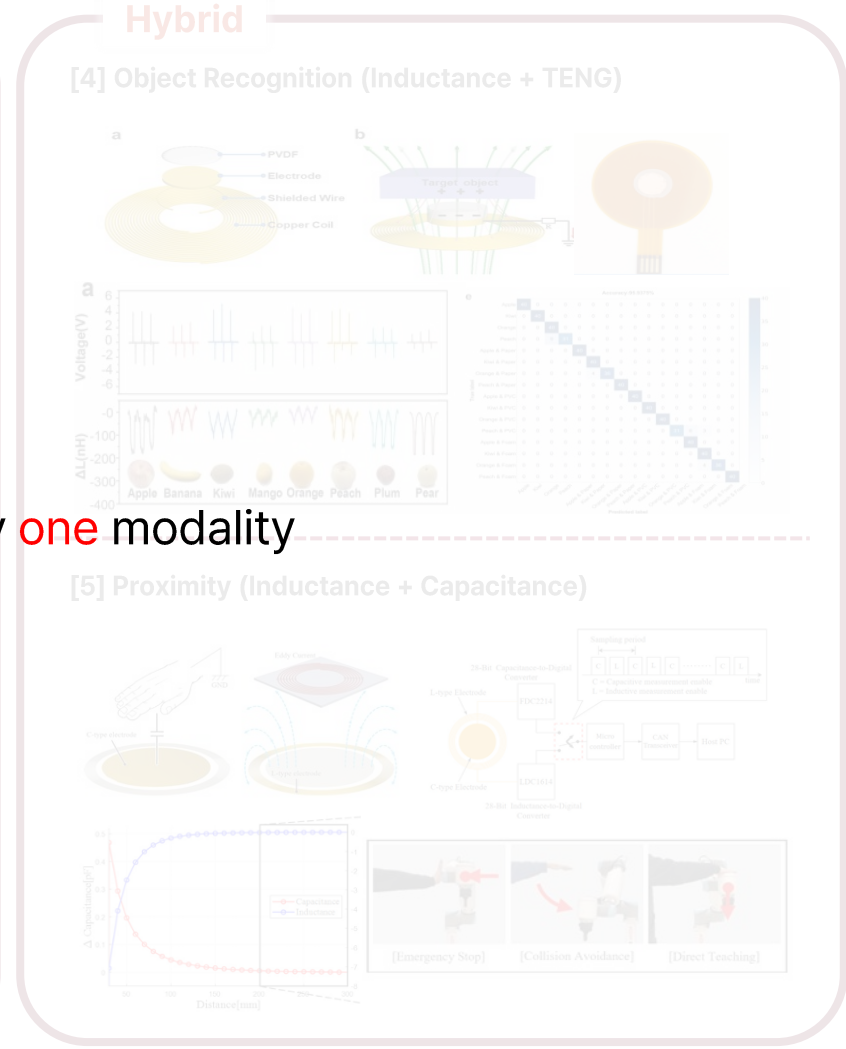
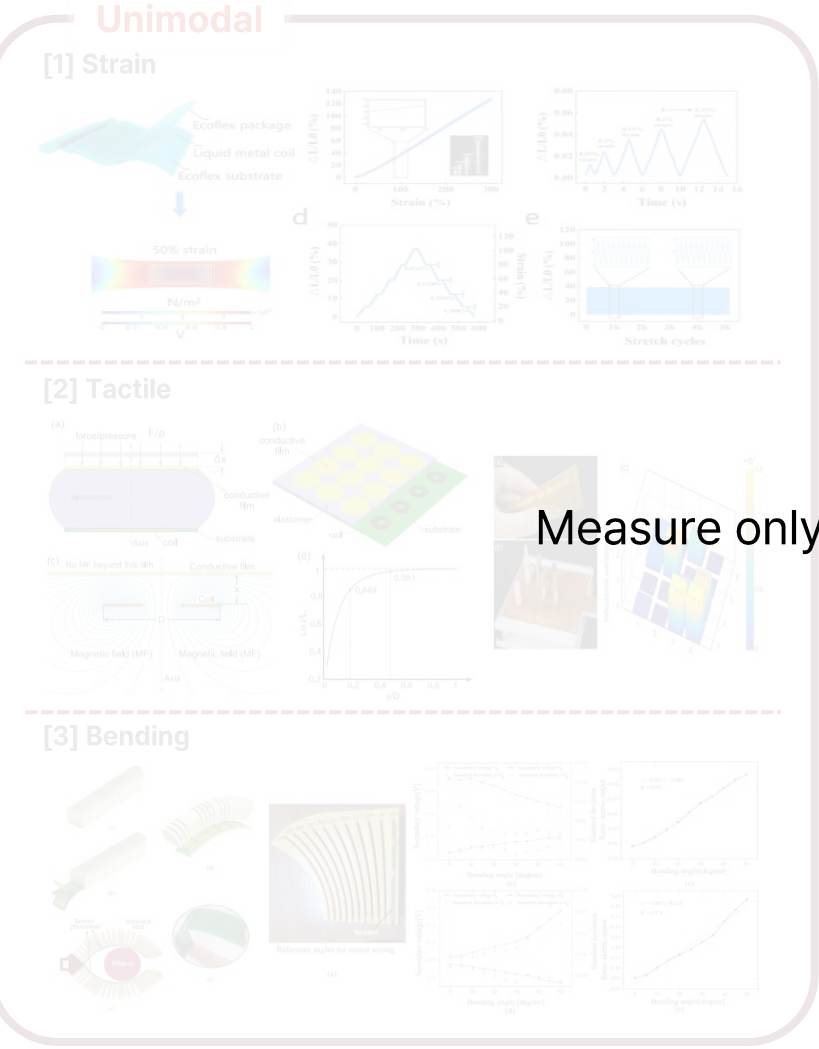


< Contact >



Introduction – Soft Inductive Sensor

Previous Researches



[1] "Ultra-high resolution, multi-scenario, super-elastic inductive strain sensors based on liquid metal for the wireless monitoring of human movement." Materials Advances (2024)

[2] "Robust and high-performance soft inductive tactile sensors based on the Eddy-current effect." Sensors and Actuators A: Physical (2018)

[3] "A planar coil-based inductive bend sensor for soft robotic applications." IEEE Sensors Journal (2024)

[4] "A triboelectric-inductive hybrid tactile sensor for highly accurate object recognition." Nano Energy (2022)

[5] "Multi-functional safety sensor coupling capacitive and inductive measurement for physical human-robot interaction." Sensors and Actuators A: Physical (2023)

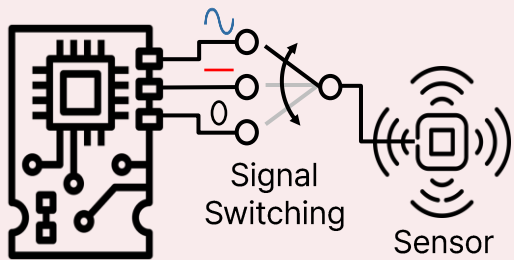
[6] "A soft inductive bimodal sensor for proprioception and tactile sensing of soft machines." Soft Robotics (2024)

[7] "Fingertip-inspired spatially anisotropic inductive liquid metal sensors with ultra-wide range, high linearity and exceptional stability." Advanced Materials (2025)

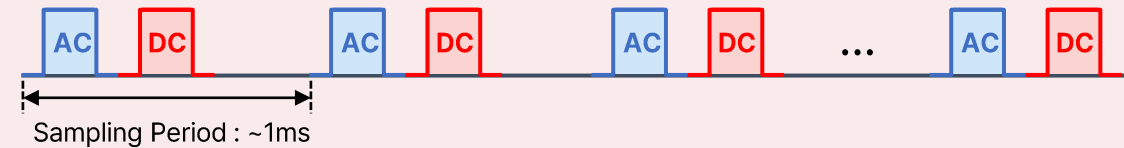
Introduction – Research Idea

Multiple Parameters Acquisition & Modalities Decoupling

Time Division Measurement(TDM)

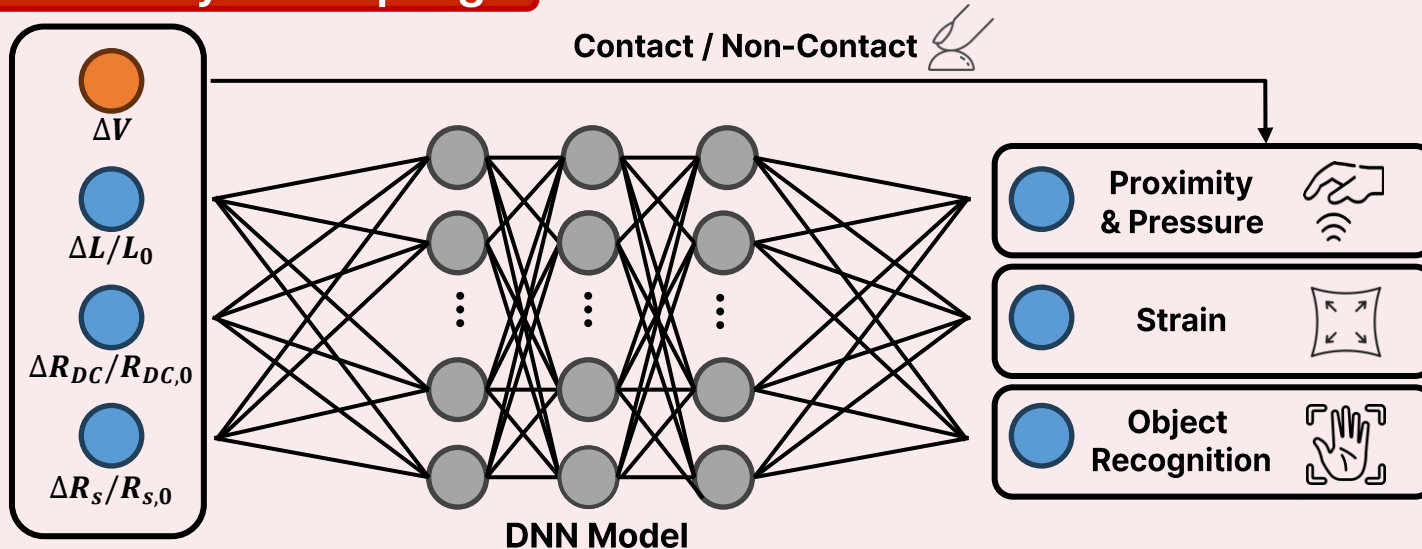


<Signal Acquisition Timeline>



AC	ΔL	→ LDC1614
	$\Delta L \text{ \& } \Delta R_s$	→ LDC1101
DC	ΔR_{DC}	→ ADC
—	$\Delta V \text{ (TENG)}$	→ ADC

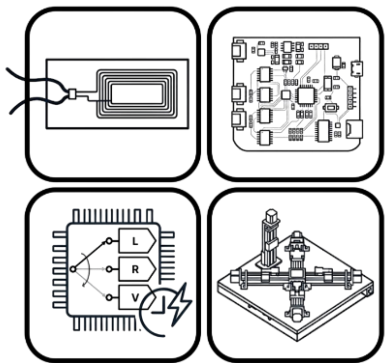
Modality Decoupling



Key Novelties

1. Overcoming Physical & Hardware Limits
 - Continuous **pre-to-post contact** spatial perception
 - **Planar** and **stretchable** True **Single-Electrode** Sensor
 - Independent Signal Extraction(expansion to **3+ modalities**)
 - Compact & Cost-Effective **Implementation**
2. Edge-AI Embedded Implementation
 - **Divergence-Free** Decoupling
 - PC-**Independent** Lightweight AI
 - Data-Efficient, Physics-Constrained **PINN**

Research Design



Part 1

Setup & Preparation

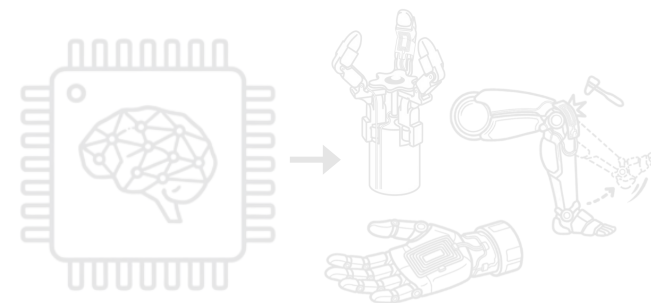
Sensor Fabrication
PCB Circuit Design
Software Development
Custom Test Platform Setup



Part 2

Data Acquisition & Learning

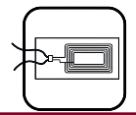
Multi-Signal Acquisition
Signal Decoupling Analysis
Dataset & Preprocessing Design
Model Training & Evaluation



Part 3

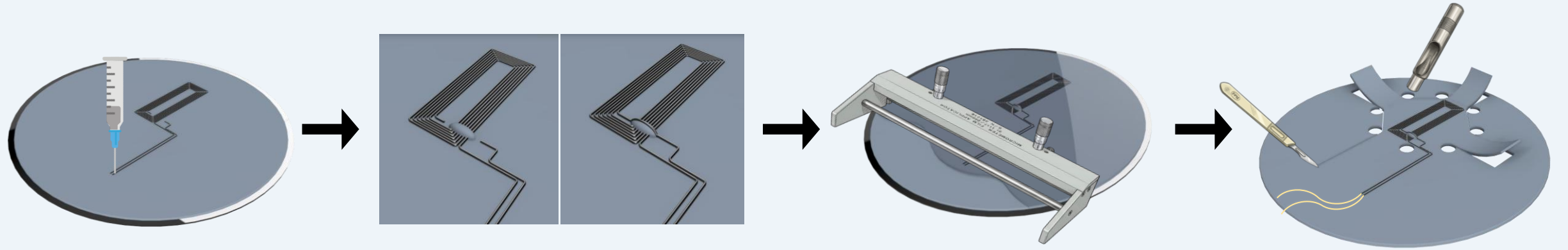
Real-time Application

Edge-AI Model Embedding
Real-time Inference System
Application scenario Design
Final System Integration



Sensor Fabrication

Sensor Fabrication Process

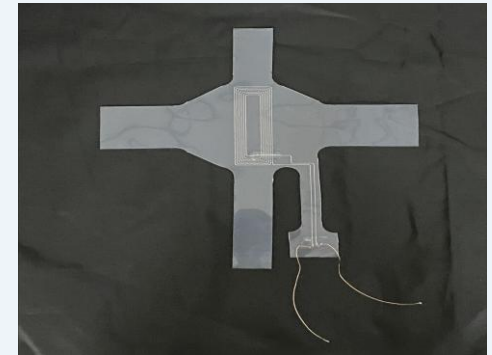
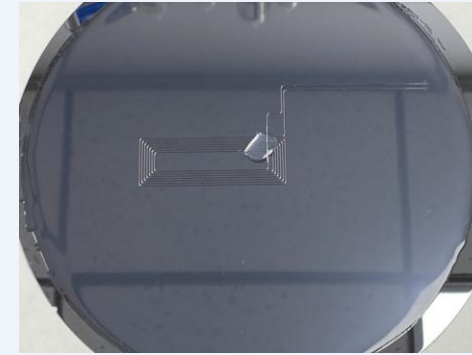
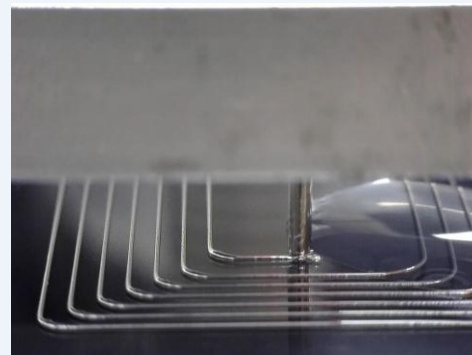
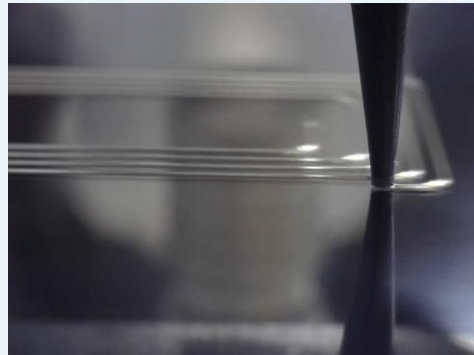


Step1
EGaIn DIW

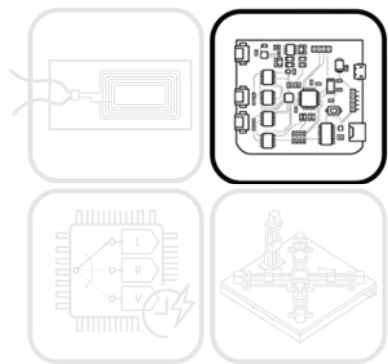
Step2
Bridging

Step3
Covering

Step4
Wire Inserting / Cutting



Research Design



Part 1

Setup & Preparation

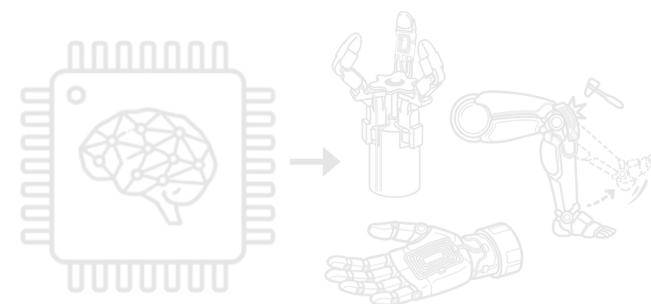
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Part 2

Data Acquisition & Learning

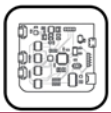
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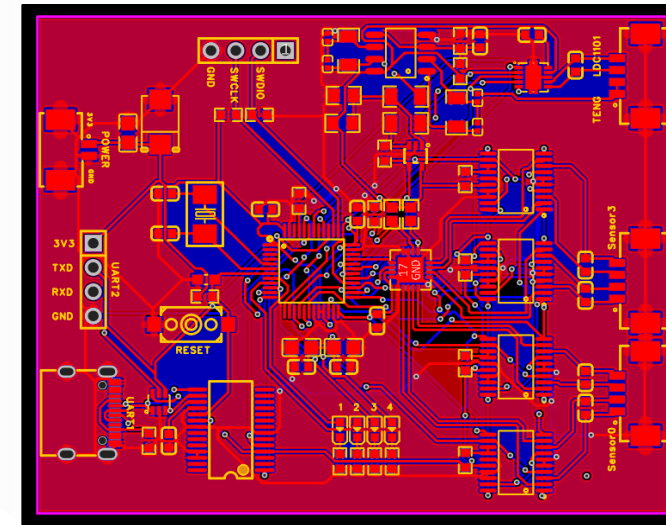
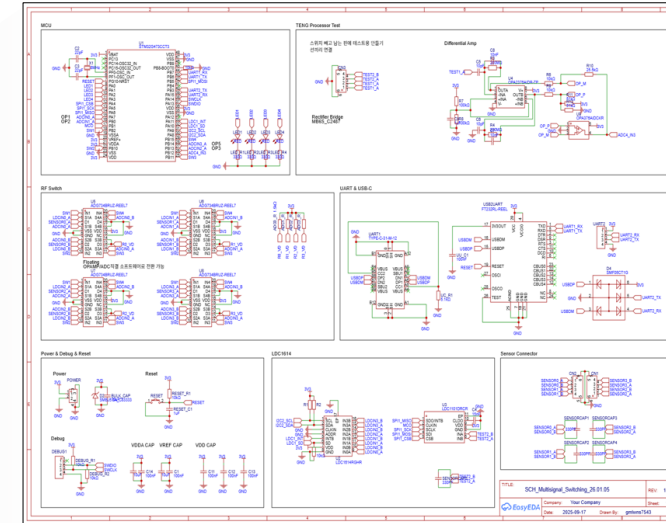
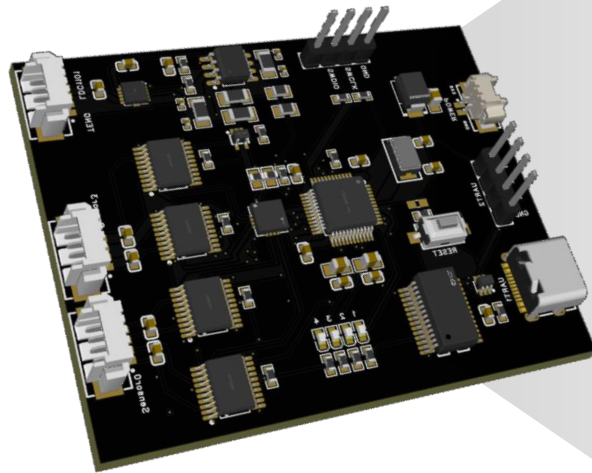
Part 3

Real-time Application

Edge-AI Model Embedding
Real-time Inference System
Application scenario Design
Final System Integration



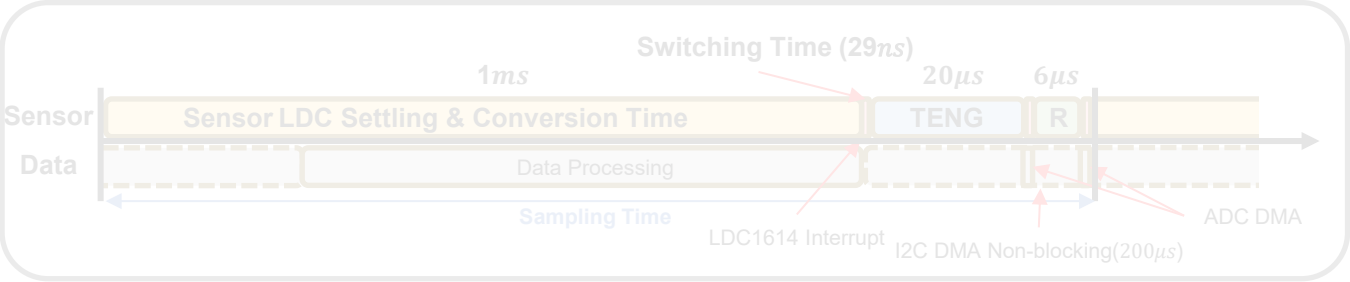
4Channel TDM Circuit



* TDM : Time Division Measurement



Circuit Working Mechanism

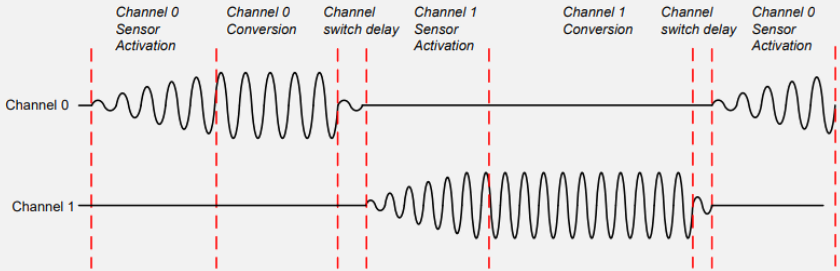


LDC 1614



- Inductance to Digital Converter
- V_{DD} : 3.3V, I_{DD} : 2.1mA, I_{drive} : < 1.5mA
- 1kHz–10MHz
- 28-bit resolution

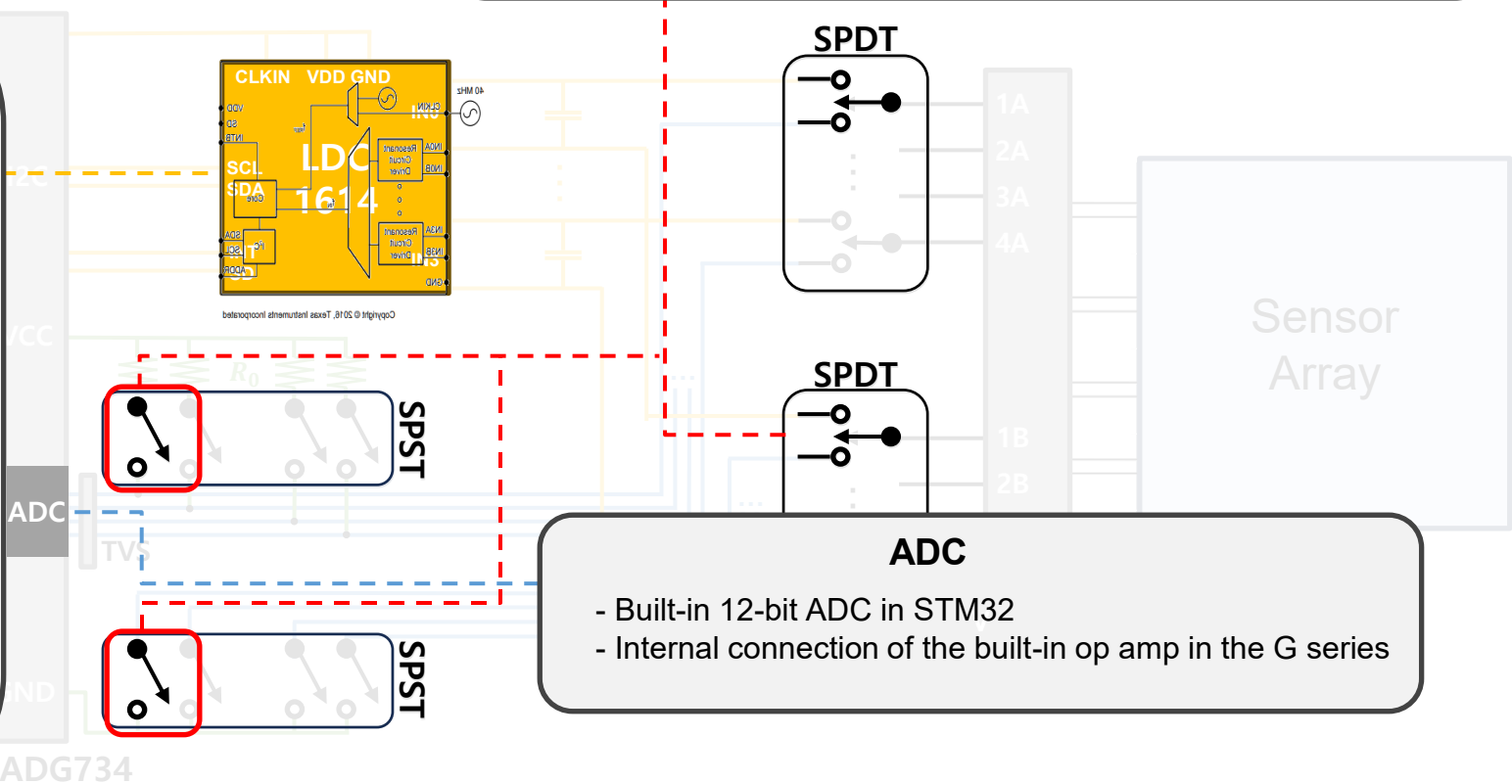
- Register Setting : $f_{in} = f_{sensor}/f_{in_divider}$, $f_{ref} = f_{clk}/f_{ref_divider}$
- $t_{settle} = (Settlecount \times 16)/f_{ref} \approx 8\mu s$,
- $t_{switch} = 692ns + 5/f_{ref} \approx 1\mu s$
- $t_{conversion} = (RCount \times 16 + 4)/f_{ref} \approx 991\mu s$ (3.2µs~26.2ms)
- $t_{I2C,400kHz} \approx 420\mu s$





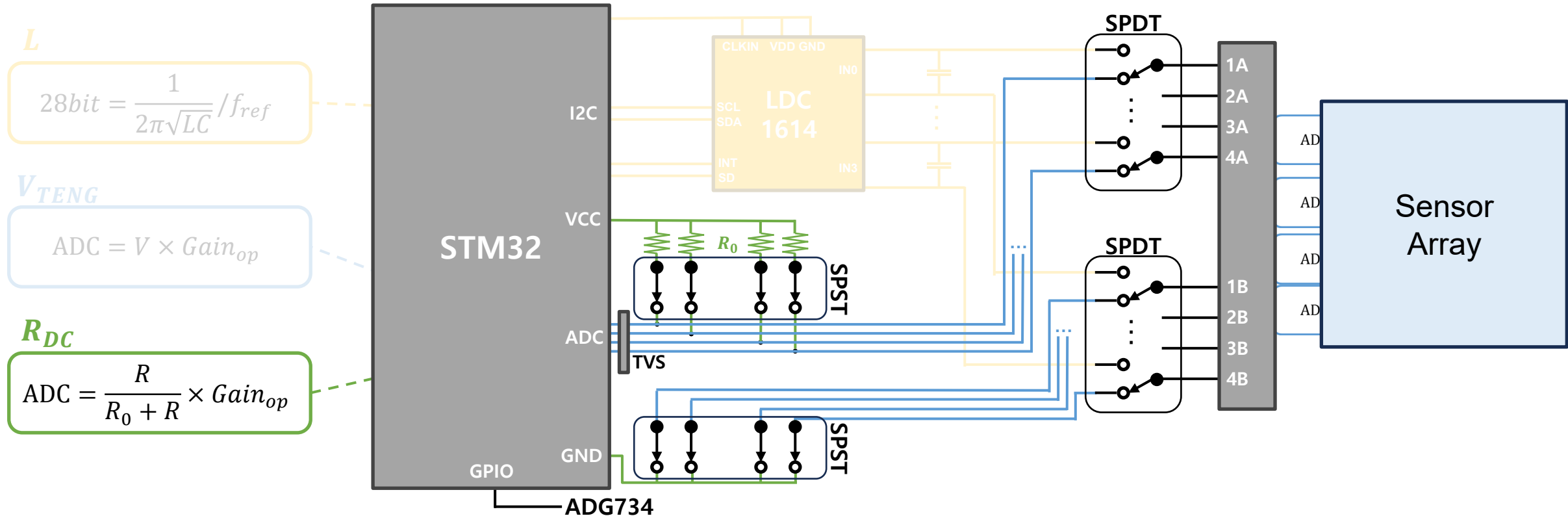
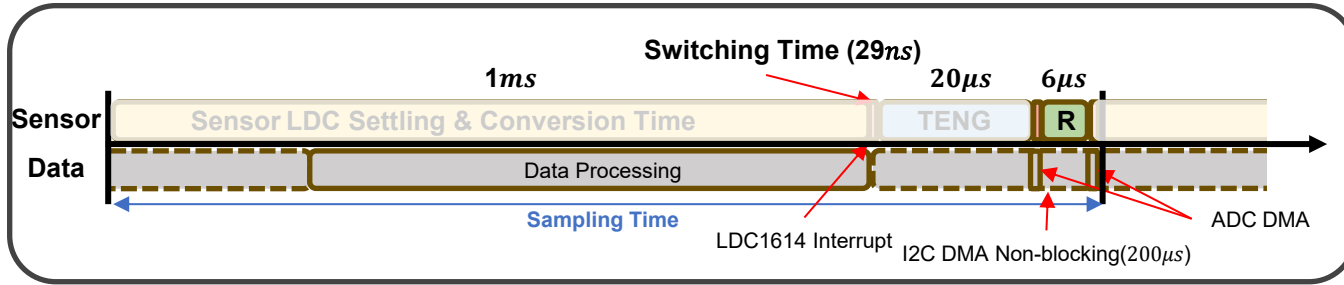
ADG734
x4

- RF-Switch (Quad SPDT)
- R_{on} : 2.5Ω, C_{on} : 34pF, t_{on} : 29ns, t_{off} : 9ns
- 200MHz Bandwidth
- : No high-frequency resonance signal loss.

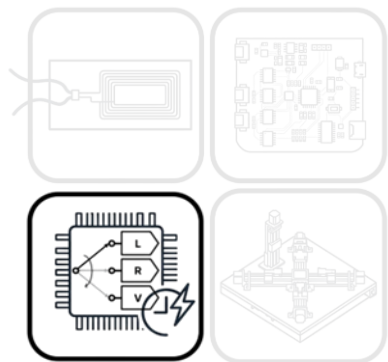




Circuit Working Mechanism



Research Design



Part 1

Setup & Preparation

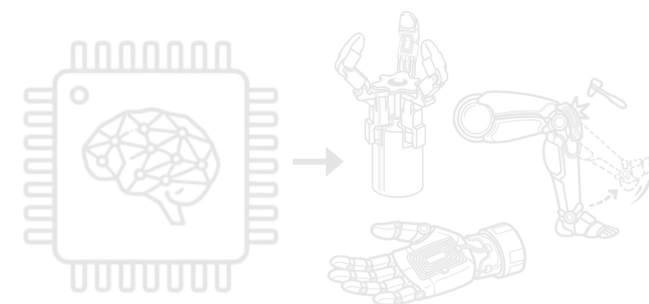
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Part 2

Data Acquisition & Learning

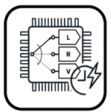
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Part 3

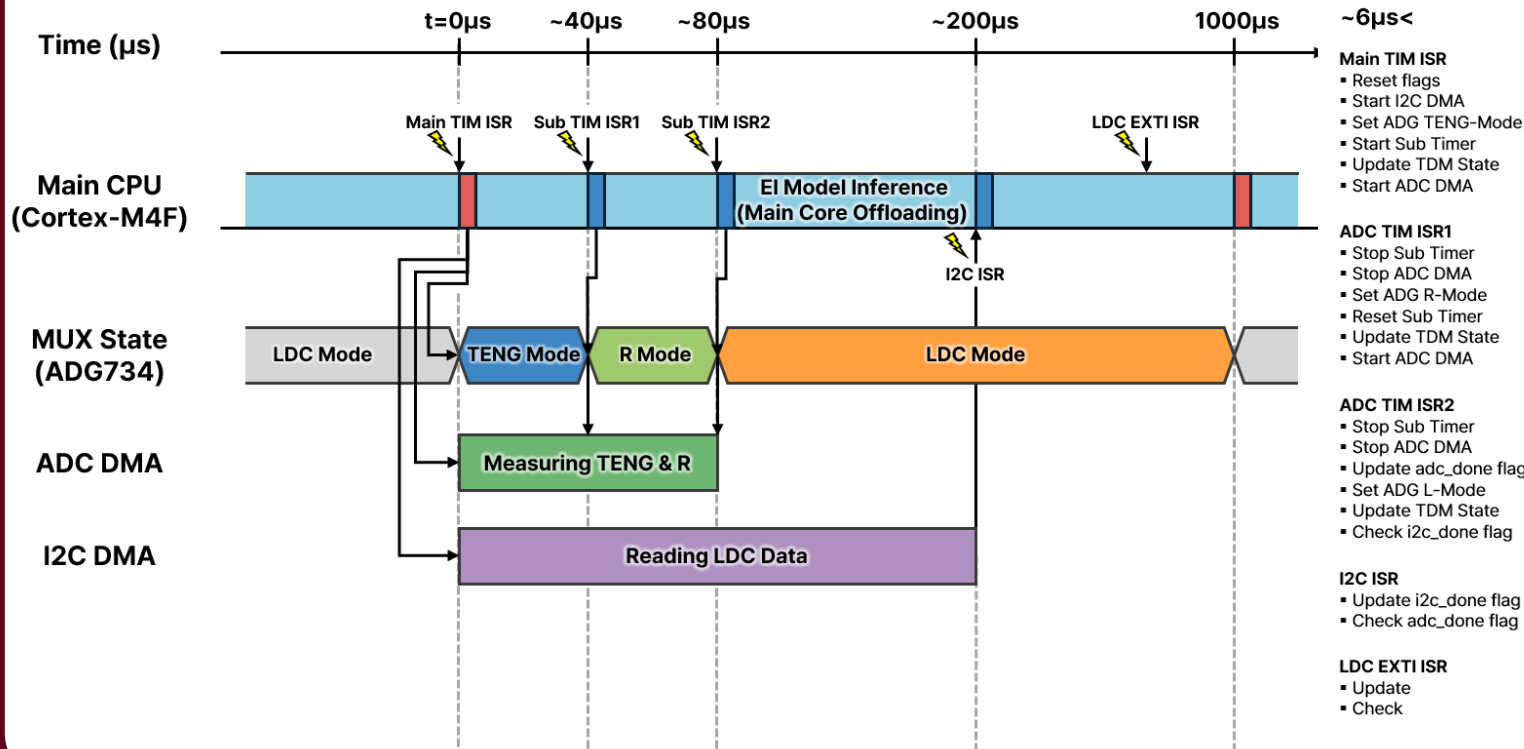
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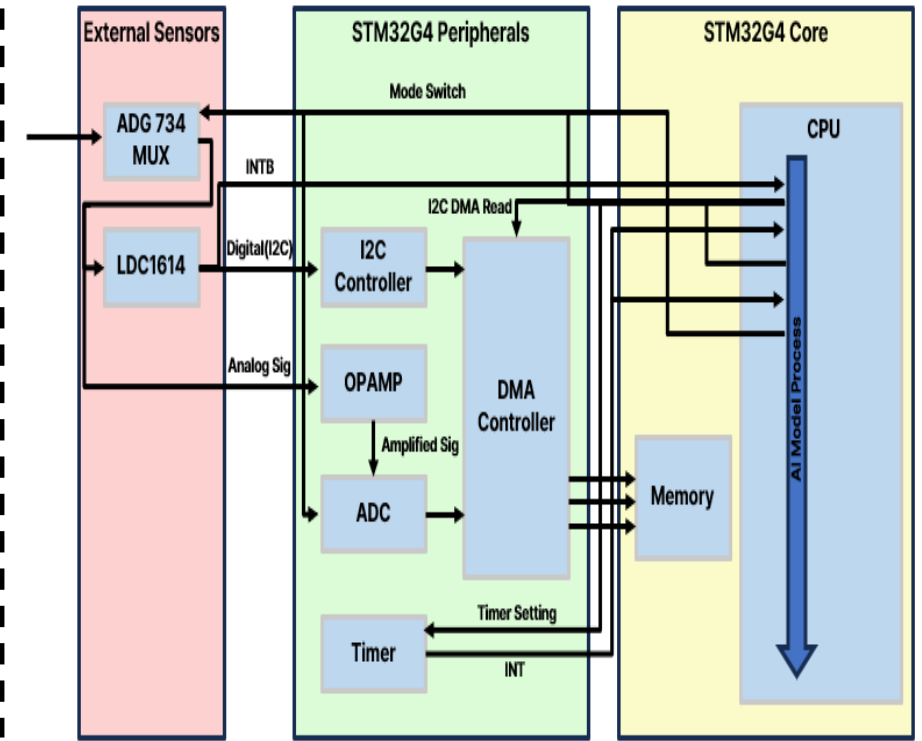


Timer Based Parallel Non-Blocking DMA Processing for Embedded AI

TDM - Timing Diagram



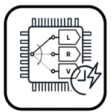
Hardware Data Flow Diagram



* TDM : Time Division Measurement

- * Sensing Period : 1kHz
- * CPU Idle Ratio : 93.908%

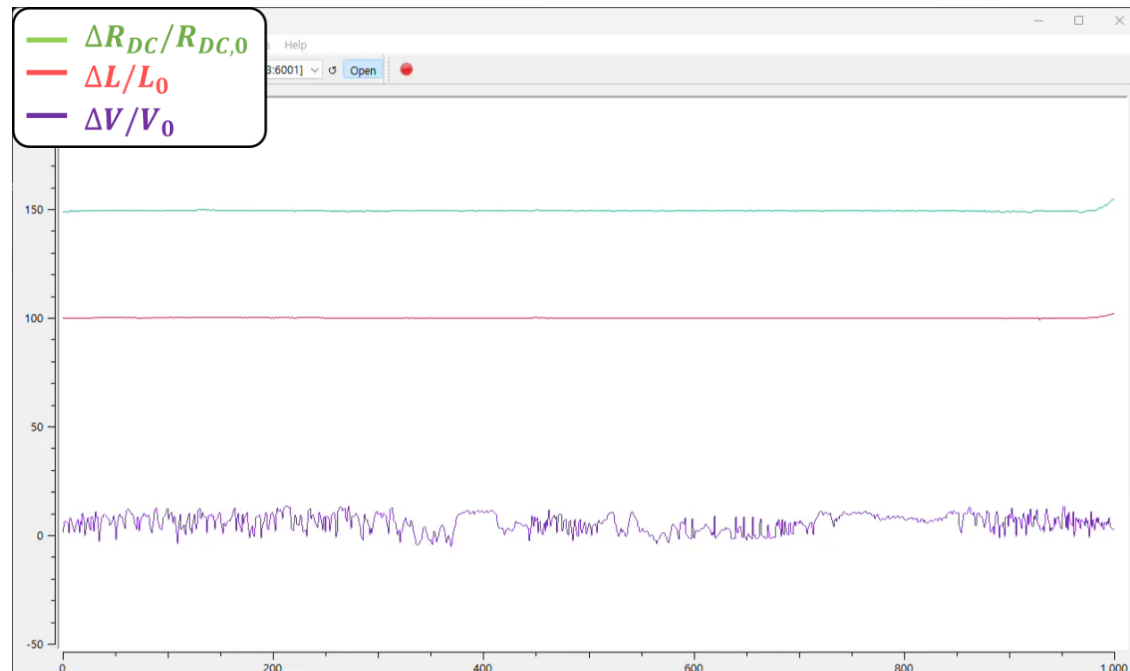
1. **Minimized overhead** : Implemented using an event-driven approach combining DMA, EXTI, and timer interrupts
2. **Fixed sampling rate** : Prevents jitter when interfacing with AI models



Results

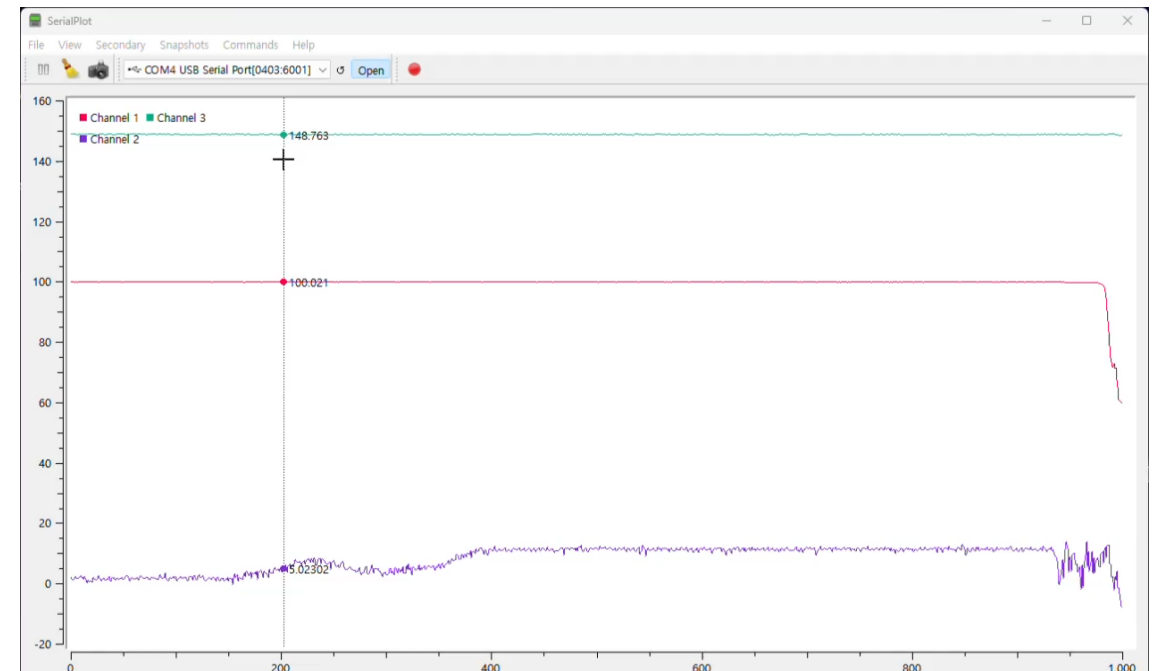
< Strain >

- Resistance and inductance increase when the sensor is stretched

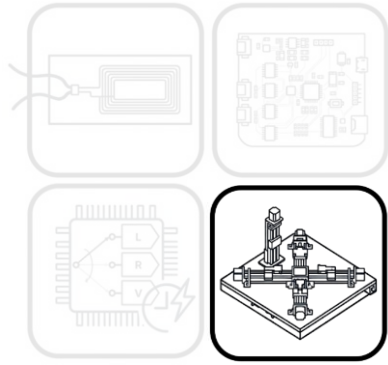


< Conductor Approach >

- Inductance decreases when a conductor approaches, and a TENG signal (peak) is generated upon contact



Research Design



Part 1

Setup & Preparation

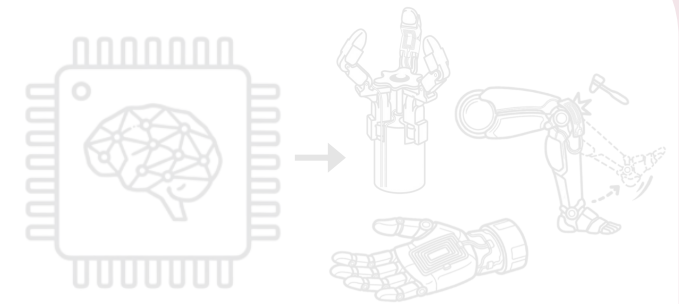
Sensor Fabrication
PCB Circuit Design
Software Development
Custom Test Platform Setup



Part 2

Data Acquisition & Learning

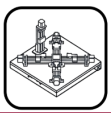
Multi-Signal Acquisition
Signal Decoupling Analysis
Dataset & Preprocessing Design
Model Training & Evaluation



Part 3

Real-time Application

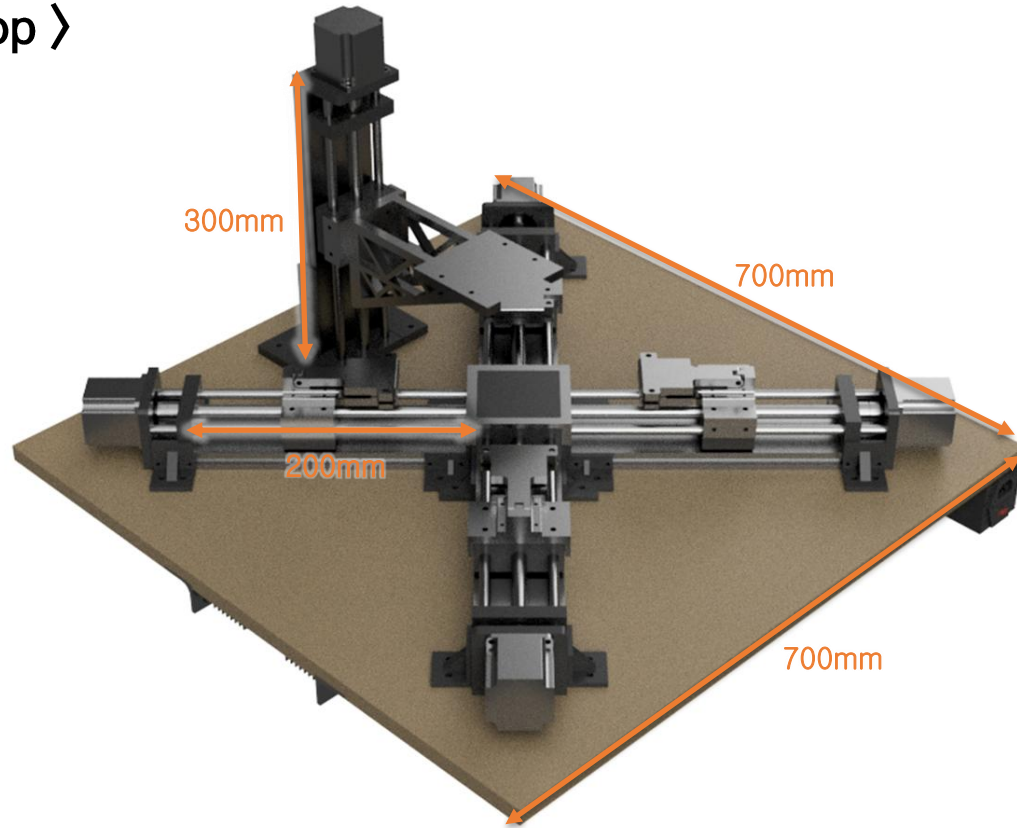
Edge-AI Model Embedding
Real-time Inference System
Application scenario Design
Final System Integration



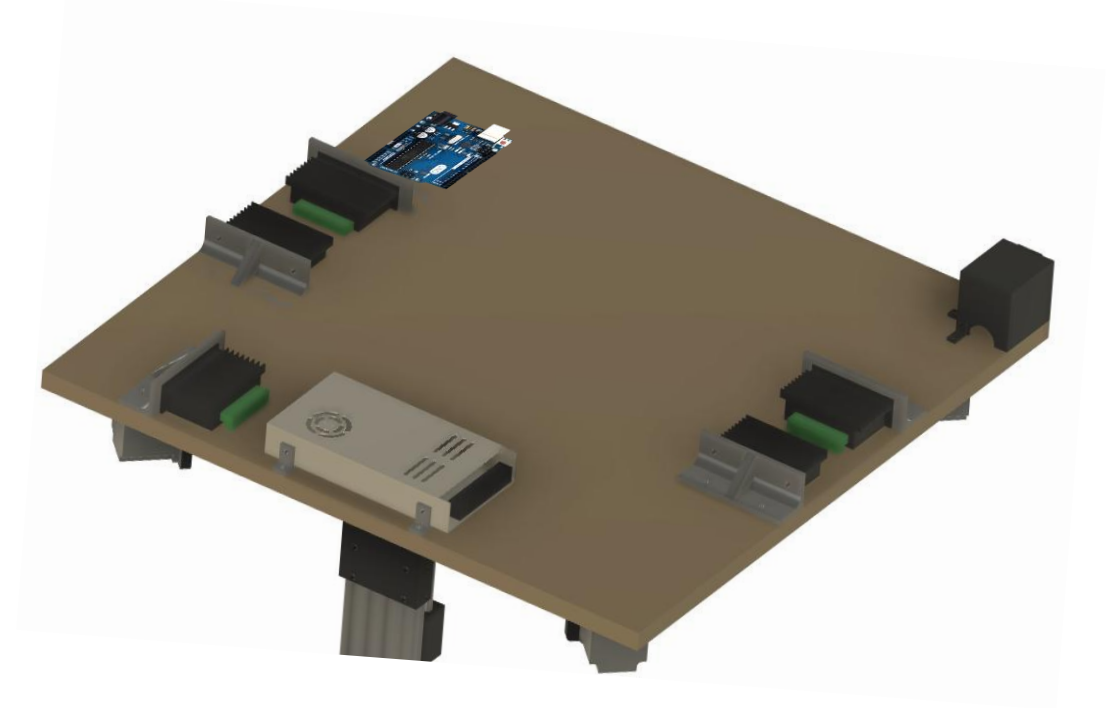
Custom Test Platform Setup

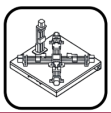
Fabrication of a test platform for data acquisition

〈 Top 〉



〈 Bottom 〉





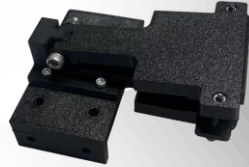
Custom Test Platform Setup

Fabrication of a test platform for data acquisition

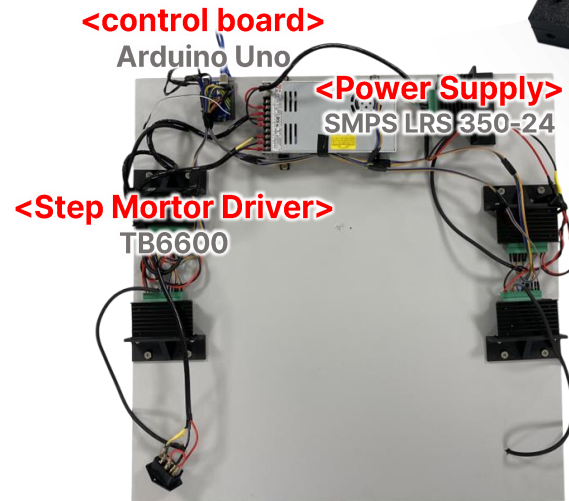
<Side>



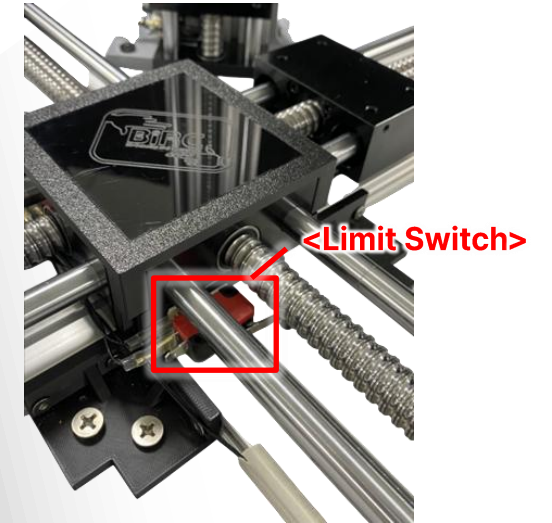
<Sensor Mounting Jig>



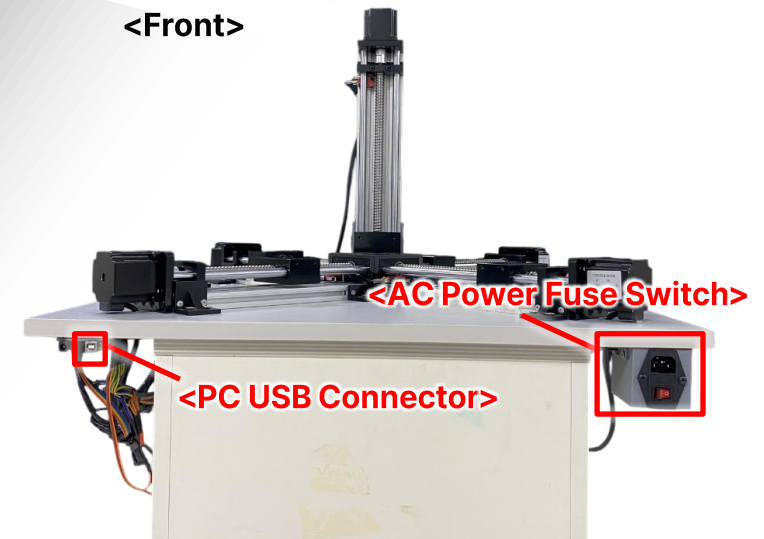
<Bottom>



<Center>



<Front>





Custom Test Platform Setup

Control program and UI development

Positioning Stage Controller V3.2

LIVE POSITION (mm): **XA: 0.000** **XB: 0.000** **YA: 0.000** **YB: 0.000** **Z: 0.000**

Serial Port: COM1 Disconnected

EMG NORMAL Motor LOCKED Settings

1. Calibration (Manual) 2. Sequence Program

Manual Mode

Manual Movement (0.1 / 1 / 5mm)

Manual Position Adjustment (mm)

Motor XA

Motor XB

Motor YA

Motor YB

Motor Z

System Setting

1. Hardware & Comm Settings

Baud Rate: 115200

Motor Microstep: 8

Screw Pitch (mm/rev): 5.0

2. Software Safety Limits

☒ Enable

Maximum Distance (mm): 50.0

Maximum Speed (mm/s): 10.0

3. Jog Button Steps (mm)

Small (<, >): 0.1

Mid (<<, >>): 1.0

Large (<<<, >>>): 5.0

Jog Speed (mm/s): 5.0

Apply

SET CURRENT POSITION AS ZERO (0.000 mm)

Home Position Setting

Motor Enable / Disable
(can be emergency-stopped and manually adjusted when disabled)

Tensile Tester V13.0 (mm Control)

LIVE POSITION (mm): **XA: 0.000** **XB: 0.000** **YA: 0.000** **YB: 0.000** **Z: 0.000**

Serial Port: COM1 Disconnected Motor LOCKED

1. Calibration (Manual) 2. Sequence Program

Program Mode

Step Parameters(distance, speed, time) Configuration

Add Sequence Step

Action: 1. Symmetric Move Total X(mm): Total Y(mm): Z(mm): Speed(mm/s): 5.0 + Add Step

Action List

#	Type	Parameters
1	Symmetric	TotX:1.155mm, TotY:2.0mm, Z:2.0mm @ 5.0mm/s
2	Individual	XA(mm):-1.0, XB(mm):-1.0, YA(mm):2.0, YB(mm):3.0, Z(mm):5.0mm @ 6.0mm/s
3	Delay	Wait 1000ms

Action Mode

1. Symmetric Move
2. Individual Move
3. Delay

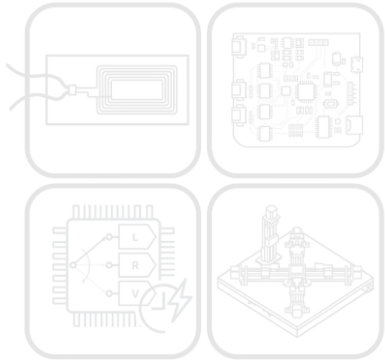
Sequence Clipboard editable (Ctrl+C/V, Delete)

Sequence Execution Button

Status Window

```
>>> START
Move: XA->0.575mm XB->-0.575mm YA->1.000mm YB->-1.000mm Z->2.000mm
Move: XA(mm)->1.000mm XB(mm)->-1.000mm YA(mm)->2.000mm YB(mm)->3.000mm Z(mm)->5.000mm
Wait 1000ms...
>>> DONE
```

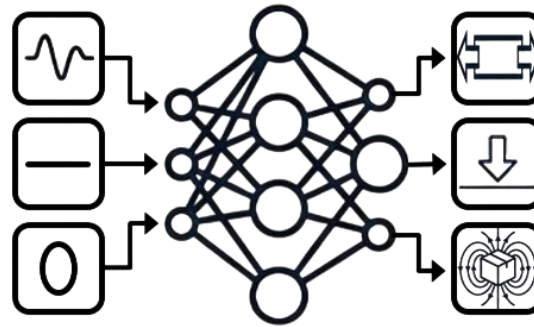
Research Design



Part 1

Setup & Preparation

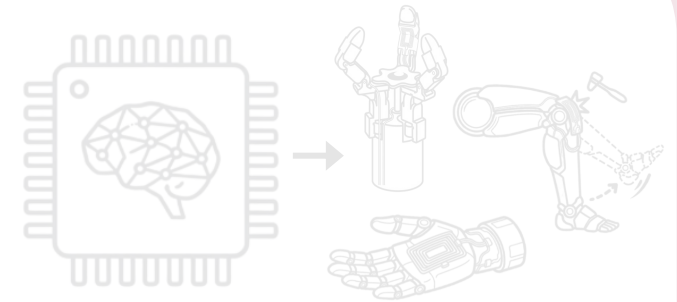
Sensor Fabrication
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Part 2

Data Acquisition & Learning

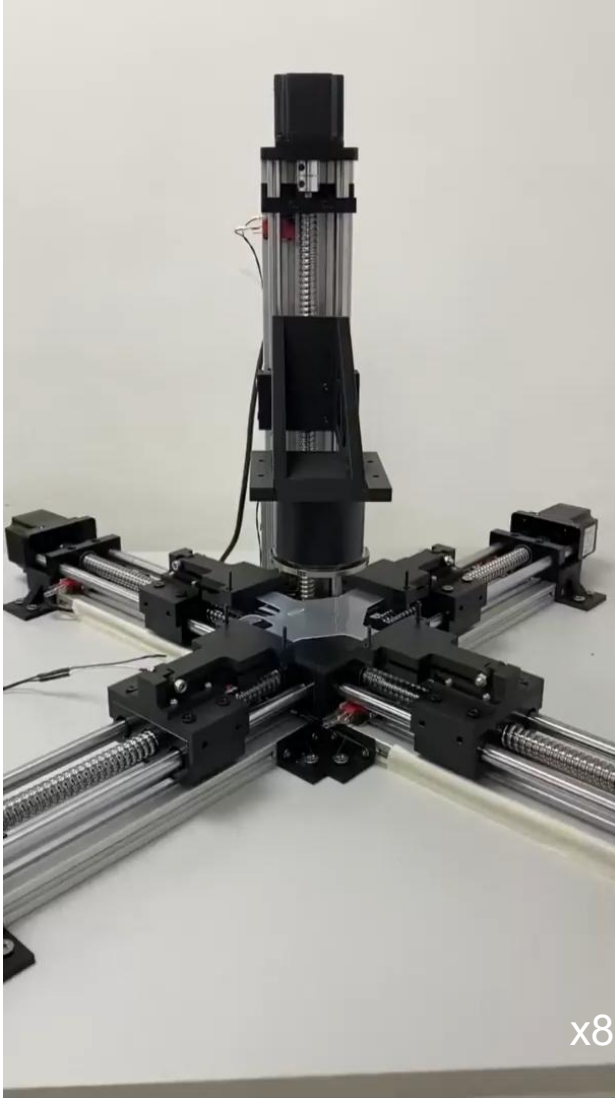
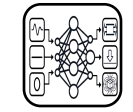
Complex Signal Acquisition
Signal Decoupling Analysis
Dataset & Preprocessing Design
Model Training & Evaluation



Part 3

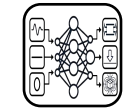
Real-time Application

Edge-AI Model Embedding
Real-time Inference System
Application scenario Design
Final System Integration



Complex Signal Experiment for Strain and Conductor Proximity

- Stretched a 120mm initial-length sensor from 0 to 36mm(30% strain) in 1mm increments (37 trials total)
- Acquired continuous data for proximity ranging from 50mm to 0mm
- Compensated for stepper motor acceleration/deceleration and synchronized spatiotemporal data axes.

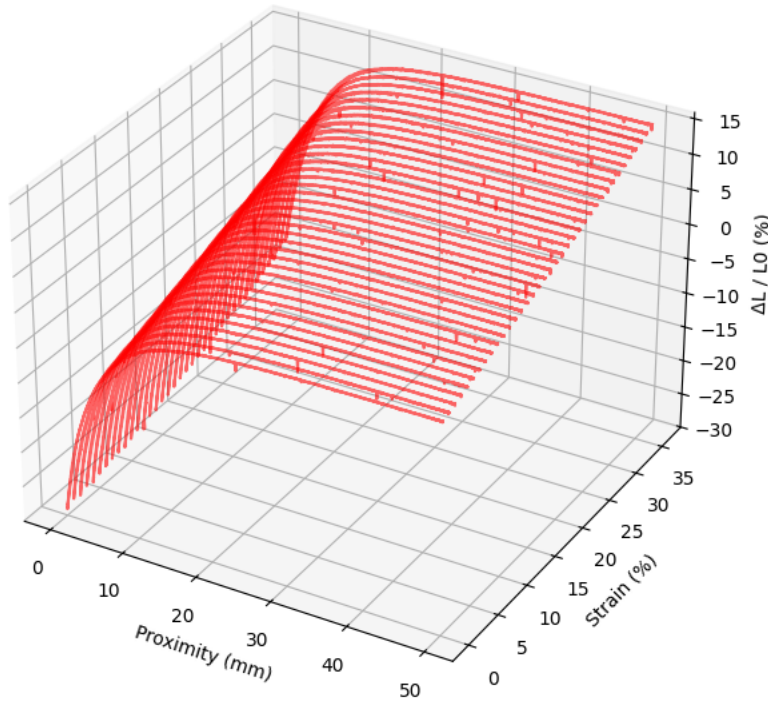


Signal Decoupling Analysis

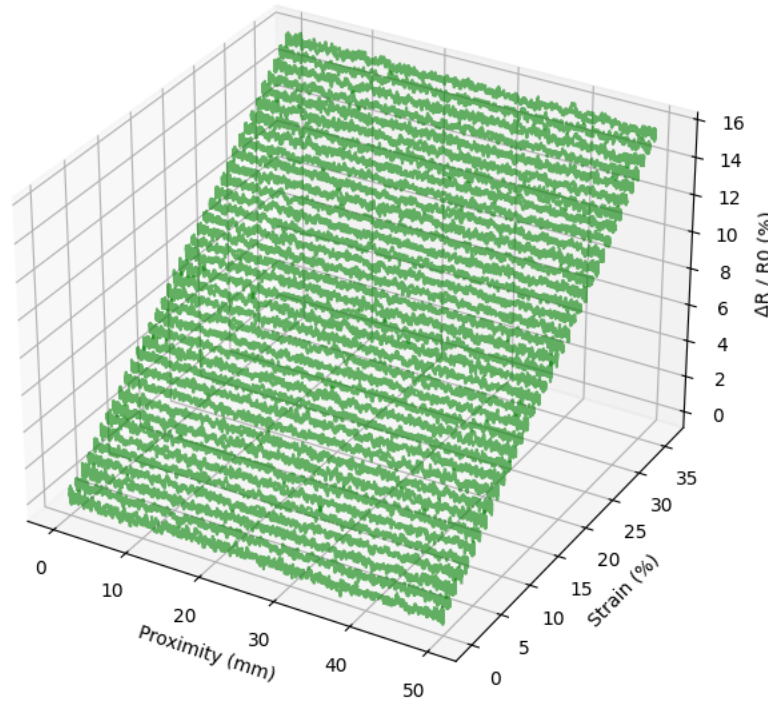
Sensor Response under Complex(Strain/Proximity) Stimuli - 3D Plot

- **Resistance:** Dominantly governed by tensile strain
- **Inductance:** Exhibits dual sensitivity to both strain and proximity

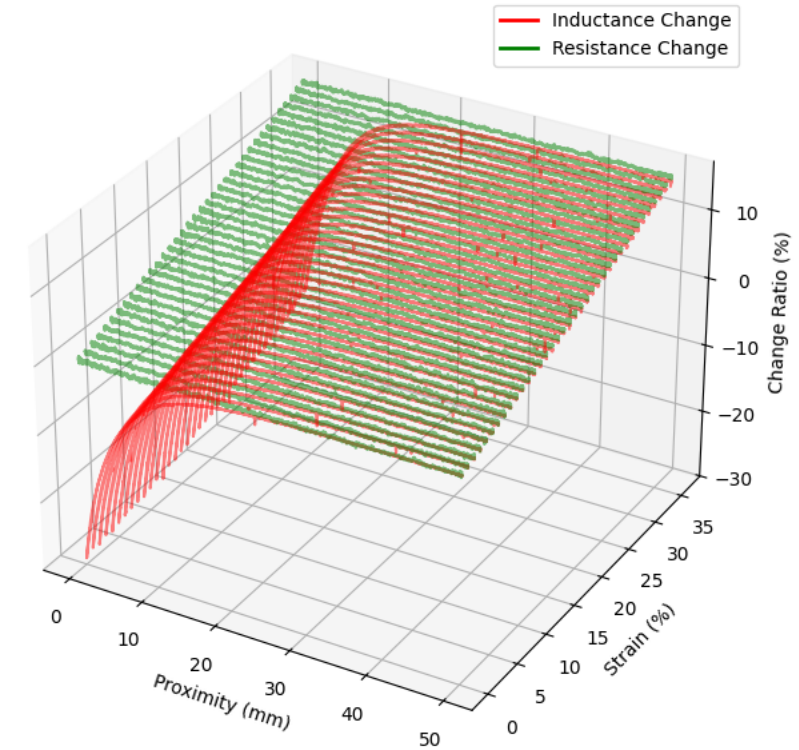
Graph 1: Inductance Change



Graph 2: Resistance Change



Graph 3: Overlapped Responses





Signal Decoupling Analysis

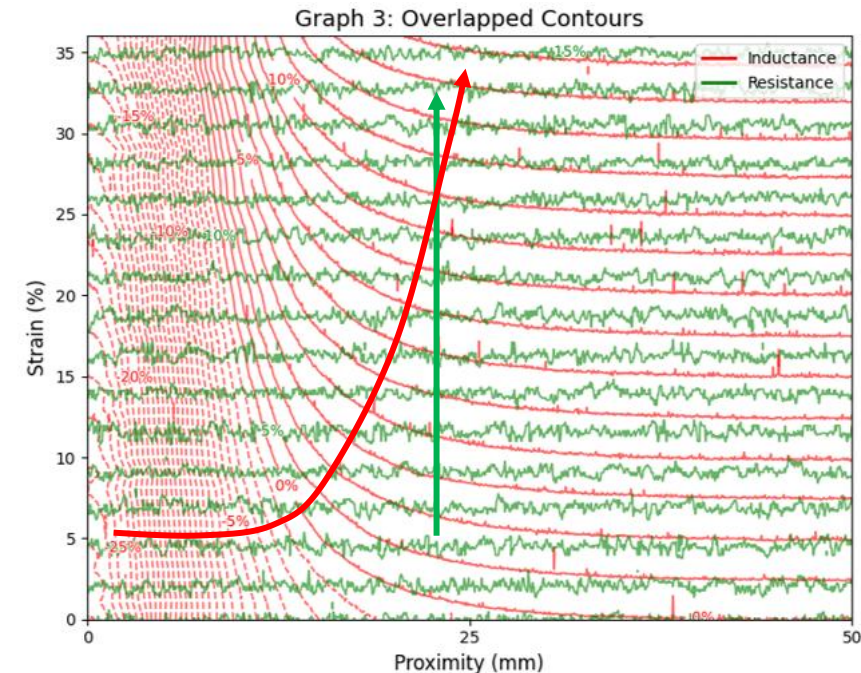
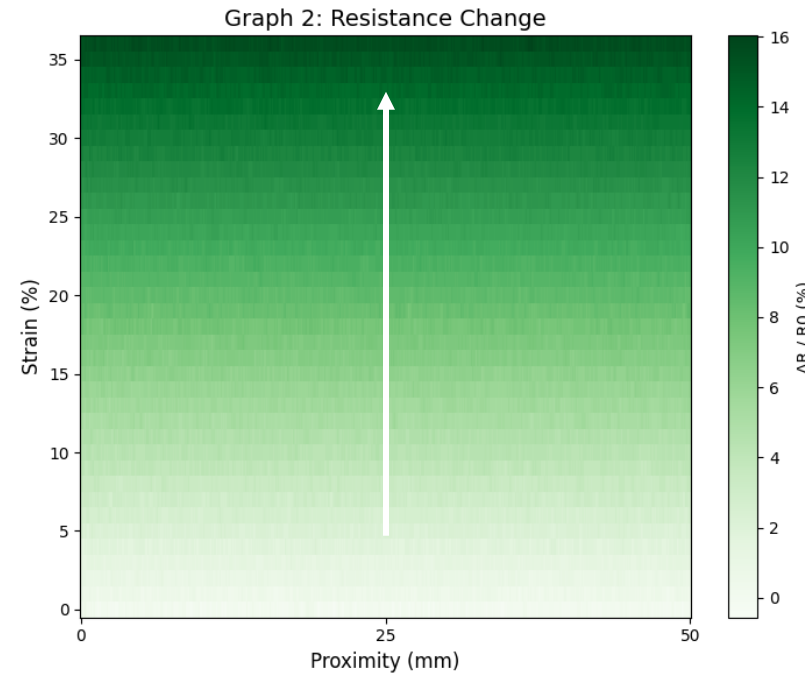
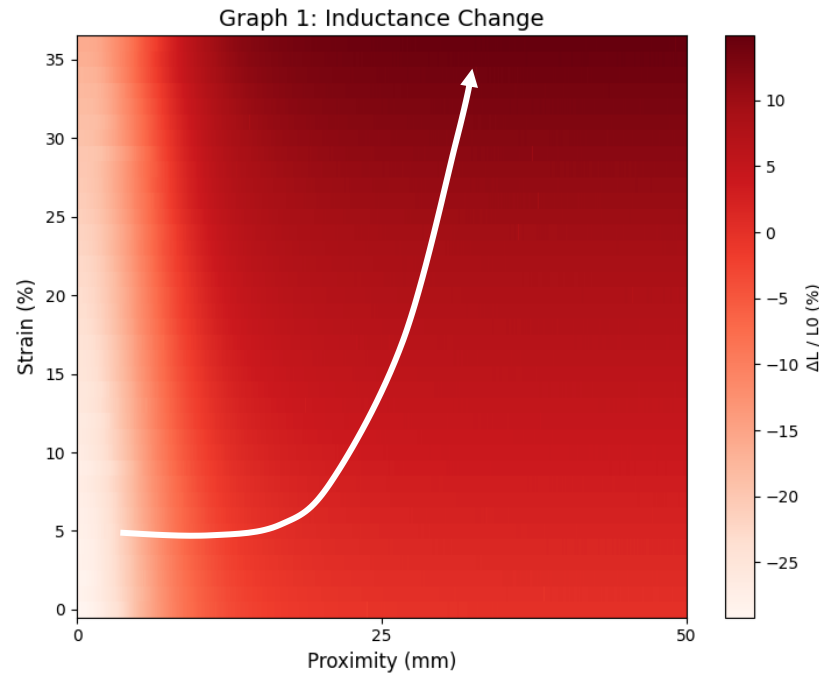
Sensor Response under Complex(Strain/Proximity) Stimuli – 2D Heatmap & Contour

- Gradients $\nabla L(d, \varepsilon)$ and $\nabla R(d, \varepsilon)$ are linearly independent
- Decoupling strain and proximity signals is feasible

$$\begin{bmatrix} \Delta R \\ \Delta L \end{bmatrix} = \begin{bmatrix} \frac{\partial R}{\partial \varepsilon} & \frac{\partial R}{\partial d} \\ \frac{\partial L}{\partial \varepsilon} & \frac{\partial L}{\partial d} \end{bmatrix} \begin{bmatrix} \Delta \varepsilon \\ \Delta d \end{bmatrix} = J \begin{bmatrix} \Delta \varepsilon \\ \Delta d \end{bmatrix}$$

gradient of R

gradient of L

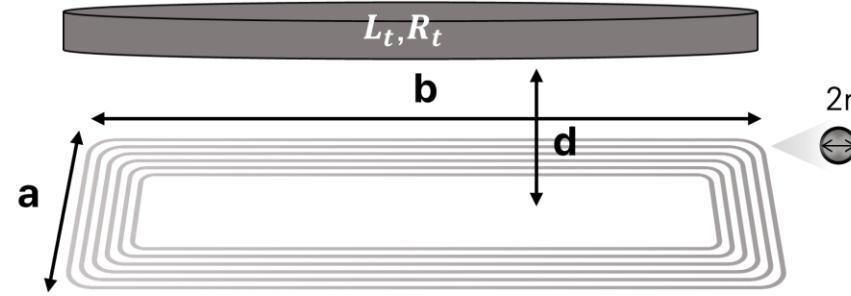
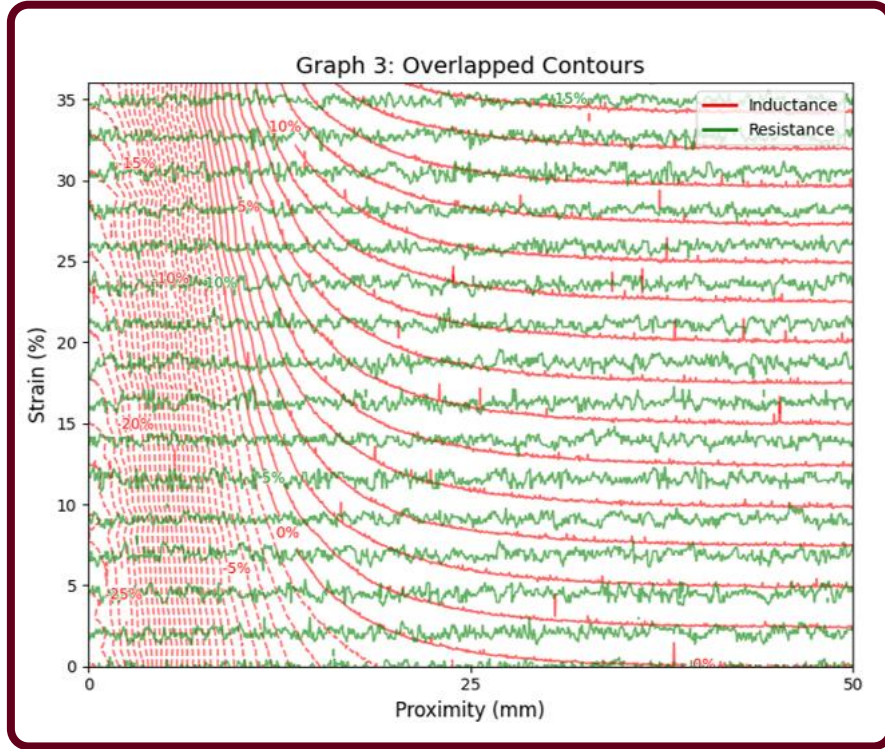


→ Is analytical inversion possible based on the governing equations?



Signal Decoupling Analysis

Sensor Response under Complex(Strain/Proximity) Stimuli – Mathematical Model



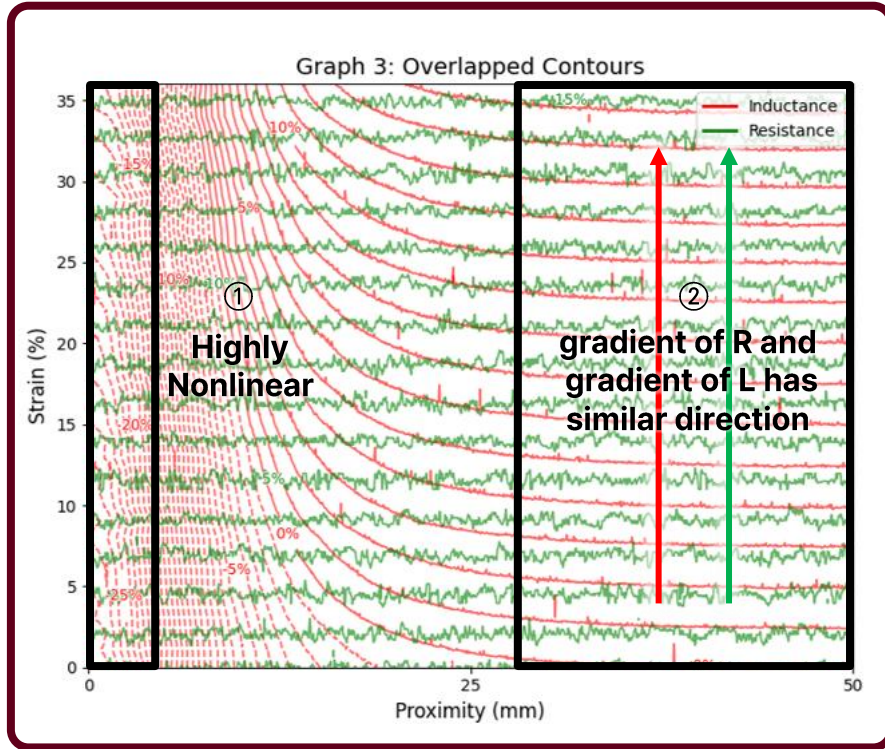
$$\begin{aligned}
 - L_{self}(\varepsilon) &= (N^2) \frac{\mu_0 \sqrt{A(\varepsilon)}}{\pi} \left[\frac{1}{\sqrt{t(\varepsilon)}} \ln \left(\frac{2\sqrt{A(\varepsilon)}}{r(\varepsilon)\sqrt{t(\varepsilon)}} \right) + \sqrt{t(\varepsilon)} \ln \left(\frac{2\sqrt{A(\varepsilon)*t(\varepsilon)}}{r(\varepsilon)} \right) + 2\sqrt{t(\varepsilon) + \frac{1}{t(\varepsilon)}} \right. \\
 &\quad \left. - \frac{1}{\sqrt{t(\varepsilon)}} \sinh^{-1} \left(\frac{1}{t(\varepsilon)} \right) - \sqrt{t(\varepsilon)} \sinh^{-1} \left(\frac{1}{t(\varepsilon)} \right) - \sqrt{t(\varepsilon)} \sinh^{-1}(t(\varepsilon)) - \frac{7}{4} \left(\frac{1}{\sqrt{t(\varepsilon)}} + \sqrt{t(\varepsilon)} \right) \right] \\
 - L(\omega, \varepsilon, d) &= L_{self}(\varepsilon) - \frac{(\omega M(\varepsilon, d))^2}{R_t^2 + (\omega L_t)^2} L_t \\
 - R_{DC}(\varepsilon) &= R_{DC,0} * (1 + \varepsilon)^2 \\
 - R_s(\omega, \varepsilon, d) &= R_{DC}(\varepsilon) * k_{skin}(\omega, r(\varepsilon)) + \frac{(\omega M(\varepsilon, d))^2}{R_t^2 + (\omega L_t)^2} R_t
 \end{aligned}$$

→ **Highly Nonlinear(Transcendental) Equation**

$$\begin{aligned}
 \frac{\Delta R}{R_0} &= \alpha_1 \varepsilon + \alpha_2 \varepsilon^2 \\
 \frac{\Delta L}{L_0} &= \beta_1 \varepsilon + \frac{\beta_2}{(d+d_0)^k} + \beta_3 \varepsilon \cdot f(d)
 \end{aligned}$$



Sensor Response under Complex(Strain/Proximity) Stimuli – Limitations of Analytical Inversion



Formulation of Linear Interpolation

< Problem ① Truncation Error in Extreme Conditions >

- Substantial errors arise from linear approximations in extreme near-field (dominated by M^2) or large strain conditions (dominated by $\sinh^{-1}, \ln$).

$$\begin{bmatrix} \varepsilon \\ d \end{bmatrix} = \begin{bmatrix} \varepsilon_0 \\ d_0 \end{bmatrix} + J^{-1} \begin{bmatrix} \Delta R \\ \Delta L \end{bmatrix} \quad \text{where} \quad J^{-1} = \frac{1}{|J|} \begin{bmatrix} \frac{\partial L}{\partial d} & -\frac{\partial R}{\partial d} \\ -\frac{\partial L}{\partial \varepsilon} & \frac{\partial R}{\partial \varepsilon} \end{bmatrix}$$

< Problem ① Truncation Error in Extreme Conditions >

- Substantial errors arise from linear approximations in extreme near-field (dominated by M^2) or large strain conditions (dominated by $\sinh^{-1}, \ln$).

< Problem ② Singularity and Loss of Orthogonality >

- Linearly dependent gradients cause the Jacobian determinant to approach zero. ($|J| \approx 0 \rightarrow J^{-1} \approx \infty$)
- This singularity leads to severe noise amplification and mathematical divergence.

→ Valid Only in Local Invertible Regions

$$\frac{\Delta R}{R_0} = \alpha_1 \varepsilon + \alpha_2 \varepsilon^2$$

$$\frac{\Delta L}{L_0} = \beta_1 \varepsilon + \frac{\beta_2}{(d+d_0)^k} + \beta_3 \varepsilon \cdot f(d)$$



Signal Decoupling Analysis

Sensor Response under Complex(Strain/Proximity) Stimuli – Necessity of Deep Neural Network (DNN)

Analytical Models

1. Mathematical Breakdown

- Jacobian singularities cause mathematical divergence and noise amplification

2. Severe Truncation Errors

- Fail in highly nonlinear regions

3. Heavy Computational Overhead

- Iterative solvers and inverse matrix calculations severely overload the MCU's limited computing power and memory

Deep Learning Model

1. Divergence-Free Inference

- Pure forward-pass operations completely eliminate singularity risks

2. Global Nonlinear Mapping

- Universally approximates complex surfaces without piecewise errors

3. Zero Overhead & $O(1)$ Latency

- Replaces heavy equations with lightweight MAC operations, guaranteeing microsecond latency on MCU

Physics Informed Neural Network(PINN) Model

$$\text{LOSS}_{\text{Total}} = \text{LOSS}_{\text{Data}} + \lambda \cdot \text{LOSS}_{\text{Physics}} \quad , \quad \text{LOSS}_{\text{Physics}} = \frac{1}{N} \sum (\|\Delta \mathbf{R} - \mathbf{R}_{\text{theory}}(\hat{\mathbf{e}})\|^2 + \|\Delta \mathbf{L} - \mathbf{L}_{\text{theory}}(\hat{\mathbf{e}}, \hat{\mathbf{d}})\|^2)$$

1. Physics-Guided Regularization

Embeds governing equations to constrain the weight space, strictly preventing non-physical predictions.

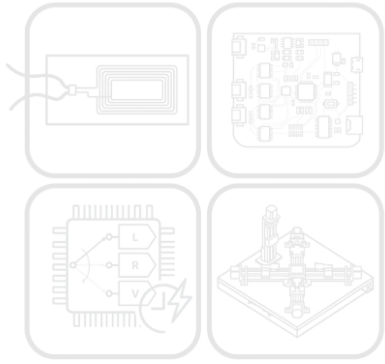
2. Robustness against OOD & Noise

Follows physical manifolds instead of wild extrapolation, perfectly preventing overfitting in unseen (Out-of-Distribution) regions.

3. Data Efficiency & Rapid Calibration

Maximizes generalization with a minimal dataset and dramatically reduces individual sensor calibration costs

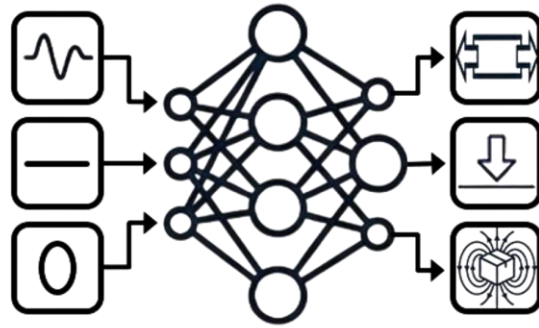
Discussion - Future Work



Part 1

Setup & Preparation

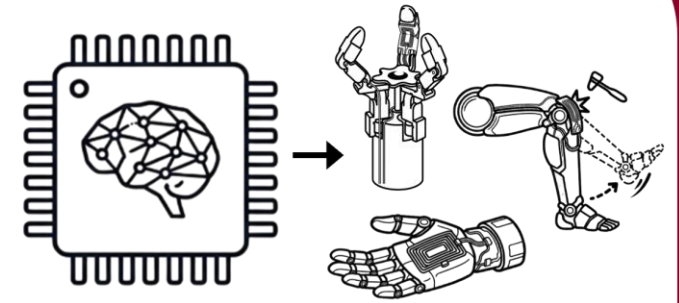
Sensor Fabrication
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Part 2

Data Acquisition & Learning

Multi-Signal Acquisition
Signal Decoupling Analysis
Dataset & Preprocessing Design
Model Training & Evaluation



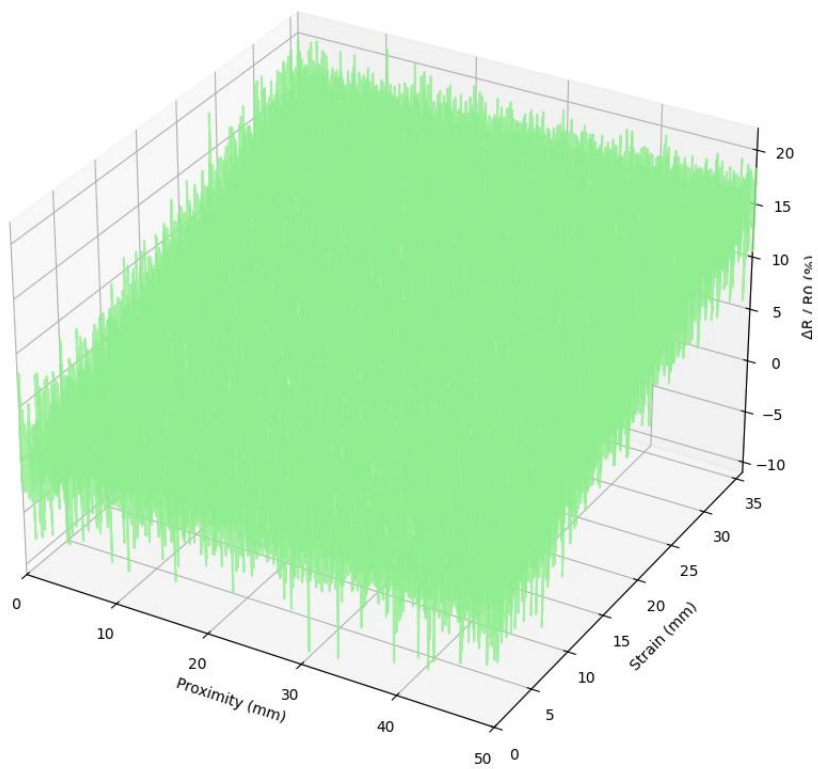
Part 3

Real-time Application

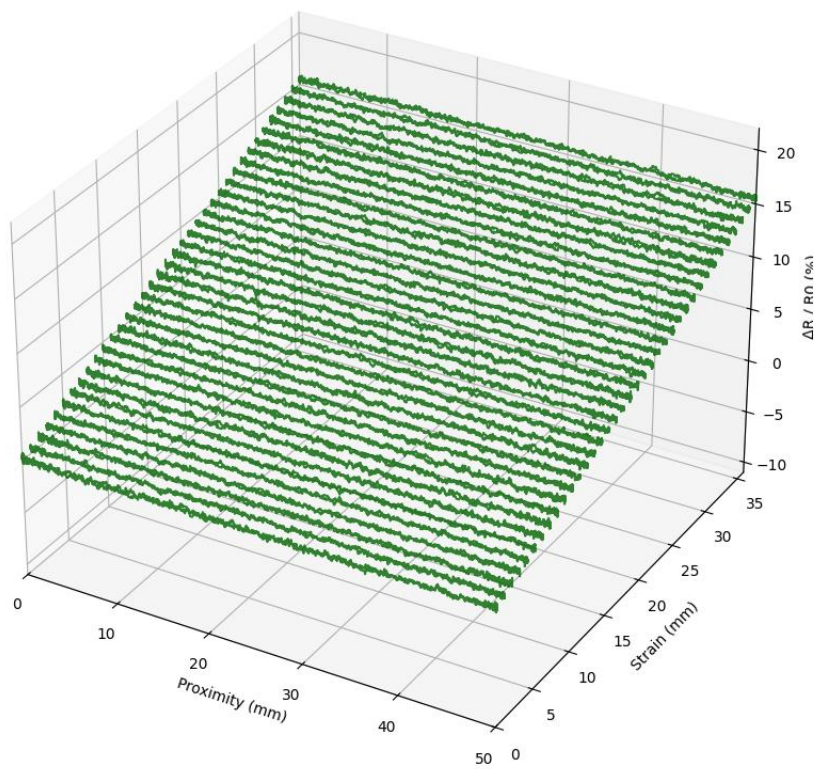
Edge-AI Model Embedding
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Appendix

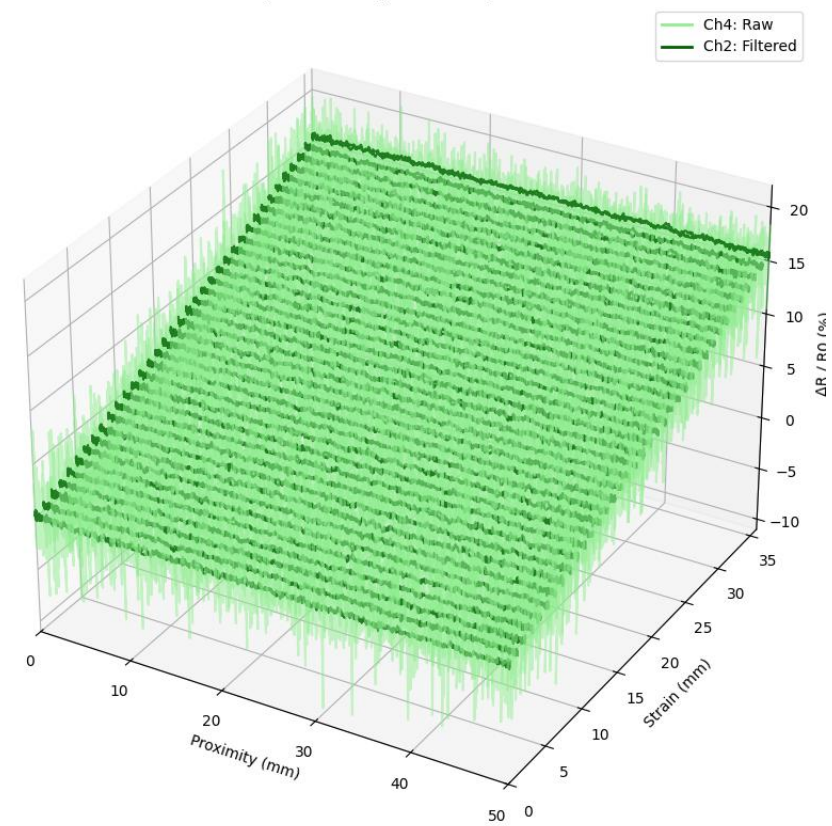
Graph 1: Channel 4 (Raw Resistance %)

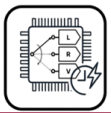


Graph 2: Channel 2 (Filtered Resistance %)



Graph 3: Filtering Effect Comparison





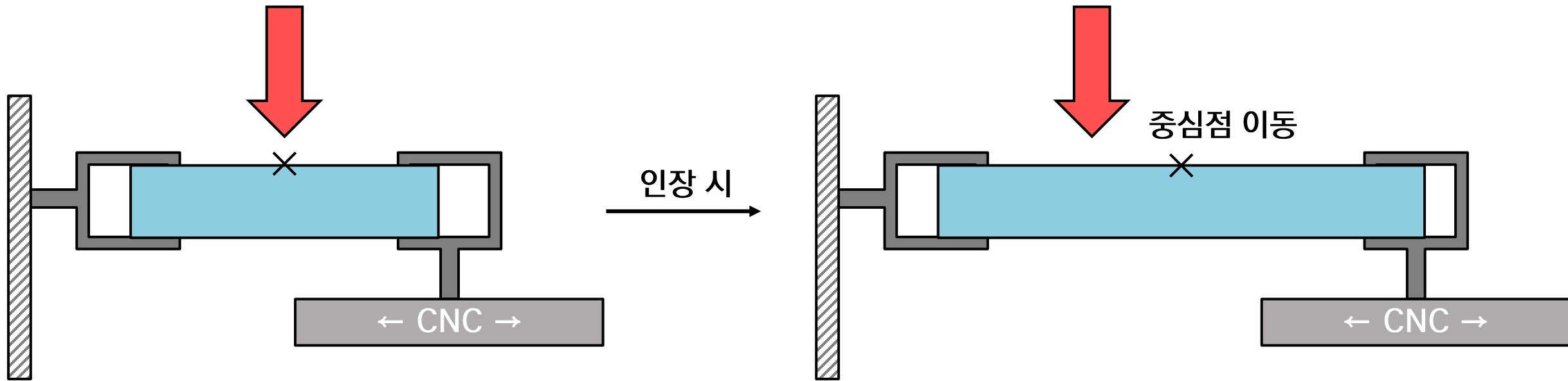
What is DMA(Direct Memory Access)?





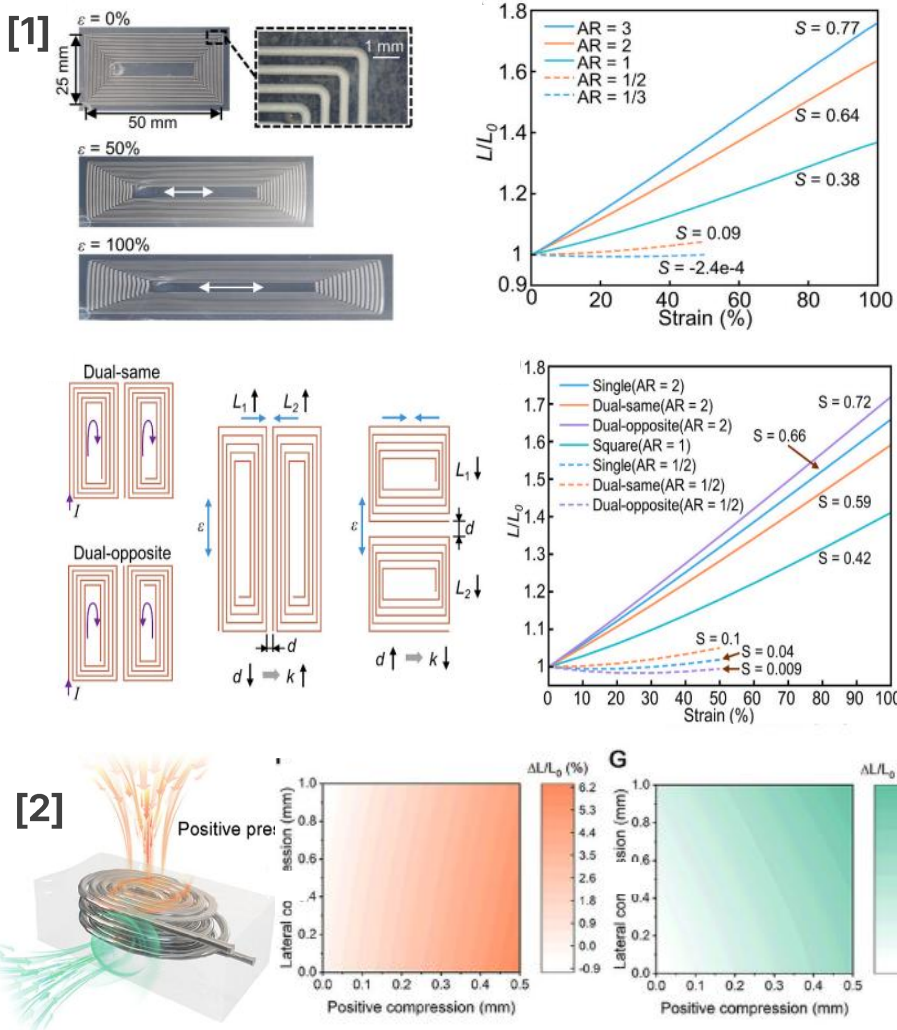
Custom Test Platform Setup

테스트 장비 제작 이유 1. 기존 CNC 머신 인장 측정 시 문제점





테스트 장비 제작 이유 2. 2D-Strain Sensing in Rectangular Spiral Coil



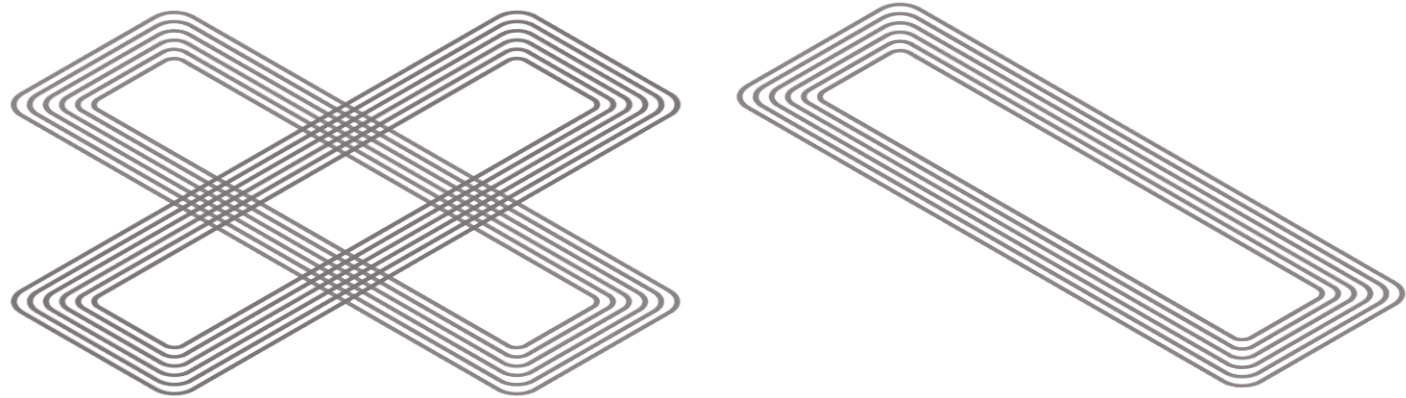
1. 종횡 길이가 다른 코일 인장의 상이한 인장 민감도

$$L_{self}(\epsilon) = (N^2) \frac{\mu_0 \sqrt{A}}{\pi} \left[\frac{1}{\sqrt{t}} \ln \left(\frac{2\sqrt{A}}{r\sqrt{t}} \right) + \sqrt{t} \ln \left(\frac{2\sqrt{A}t}{r} \right) + 2 \sqrt{t + \frac{1}{t}} - \frac{1}{\sqrt{t}} \sinh^{-1} \left(\frac{1}{t} \right) - \sqrt{t} \sinh^{-1} \left(\frac{1}{t} \right) - \sqrt{t} \sinh^{-1}(t) - \frac{7}{4} \left(\frac{1}{\sqrt{t}} + \sqrt{t} \right) \right]$$

2. L과 R은 인장에 대해 서로 다른 고유 응답 방향(effective eigen-directions)을 가지므로, 2축 인장 성분을 분해 가능

$$\begin{aligned} \Delta L &= f(\Delta l, \Delta A, \Delta d) \\ \Delta R &= f(\Delta l) \end{aligned} \quad \begin{bmatrix} \Delta L \\ \Delta R \end{bmatrix} = \begin{bmatrix} \frac{\partial L}{\partial \epsilon_x} & \frac{\partial L}{\partial \epsilon_y} \\ \frac{\partial R}{\partial \epsilon_x} & \frac{\partial R}{\partial \epsilon_y} \end{bmatrix} \begin{bmatrix} \epsilon_x \\ \epsilon_y \end{bmatrix}$$

* l : 코일 전체 길이, A : 코일 면적, d : 코일 간격



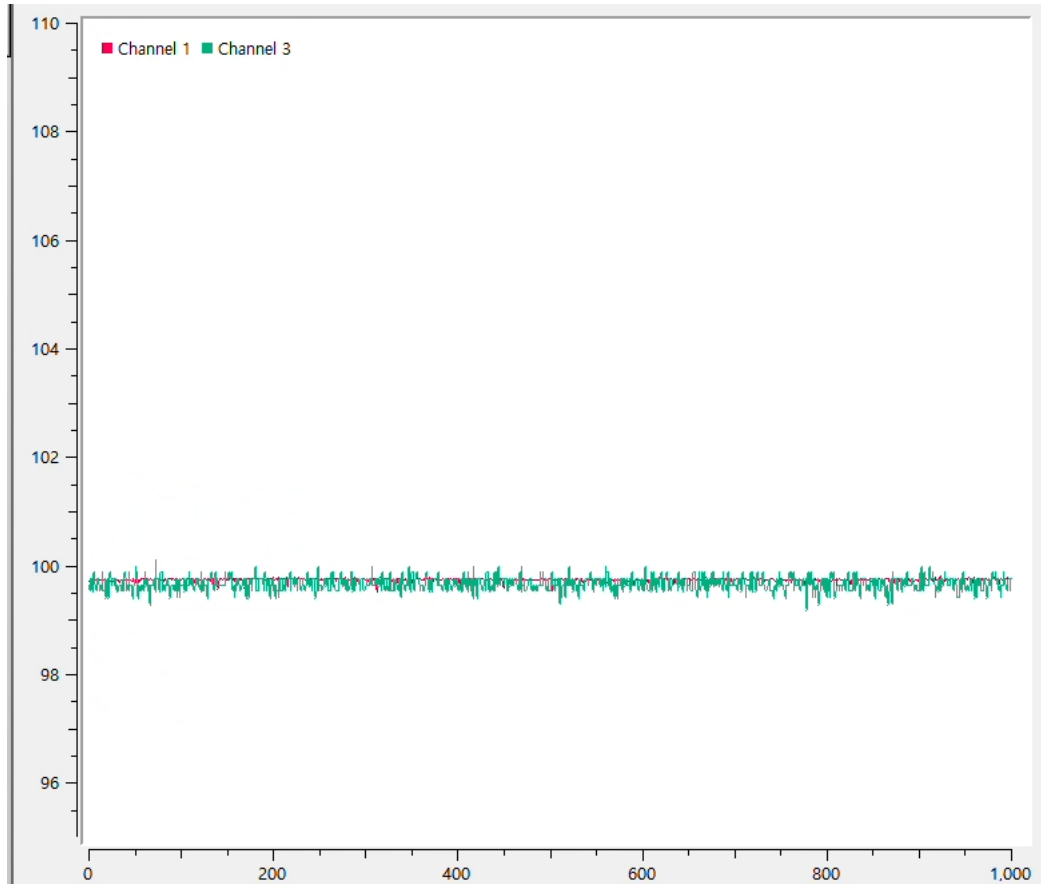


3-Axis Positioning Stage

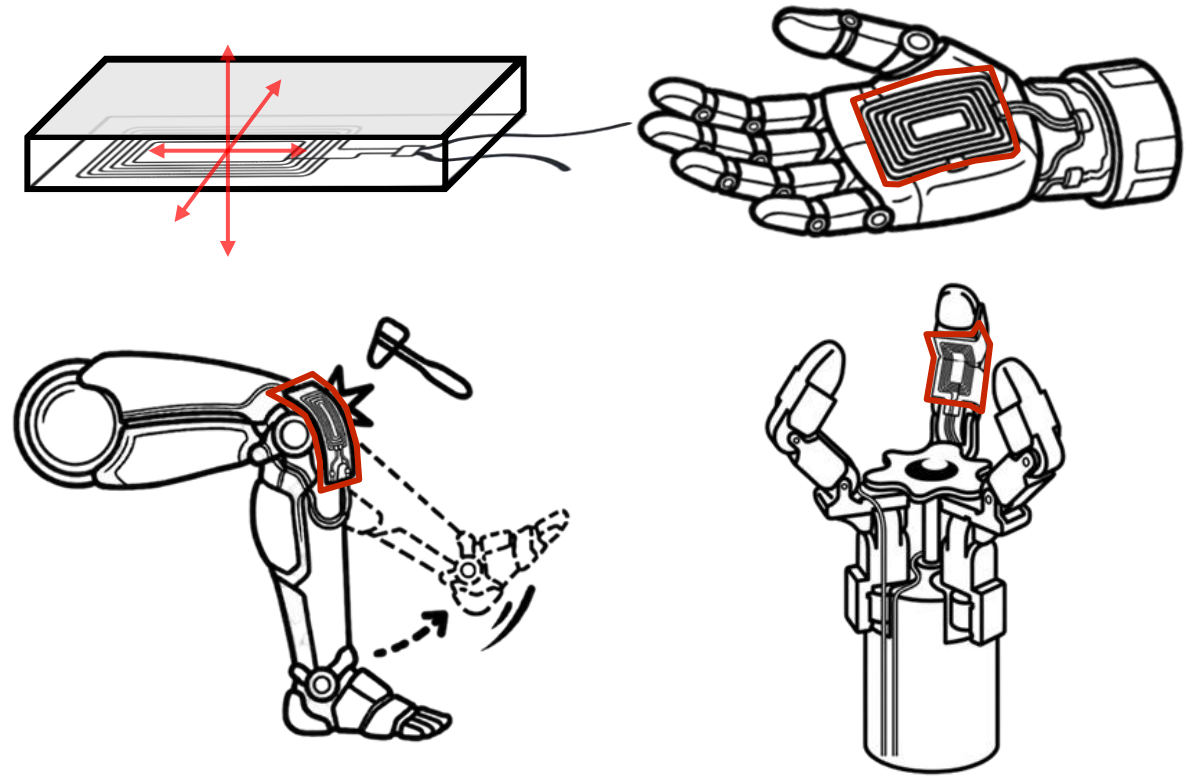
BioRobotics & Control Lab (BiRC)
Department of Mechanical Engineering
Korea University

HeeJun Moon

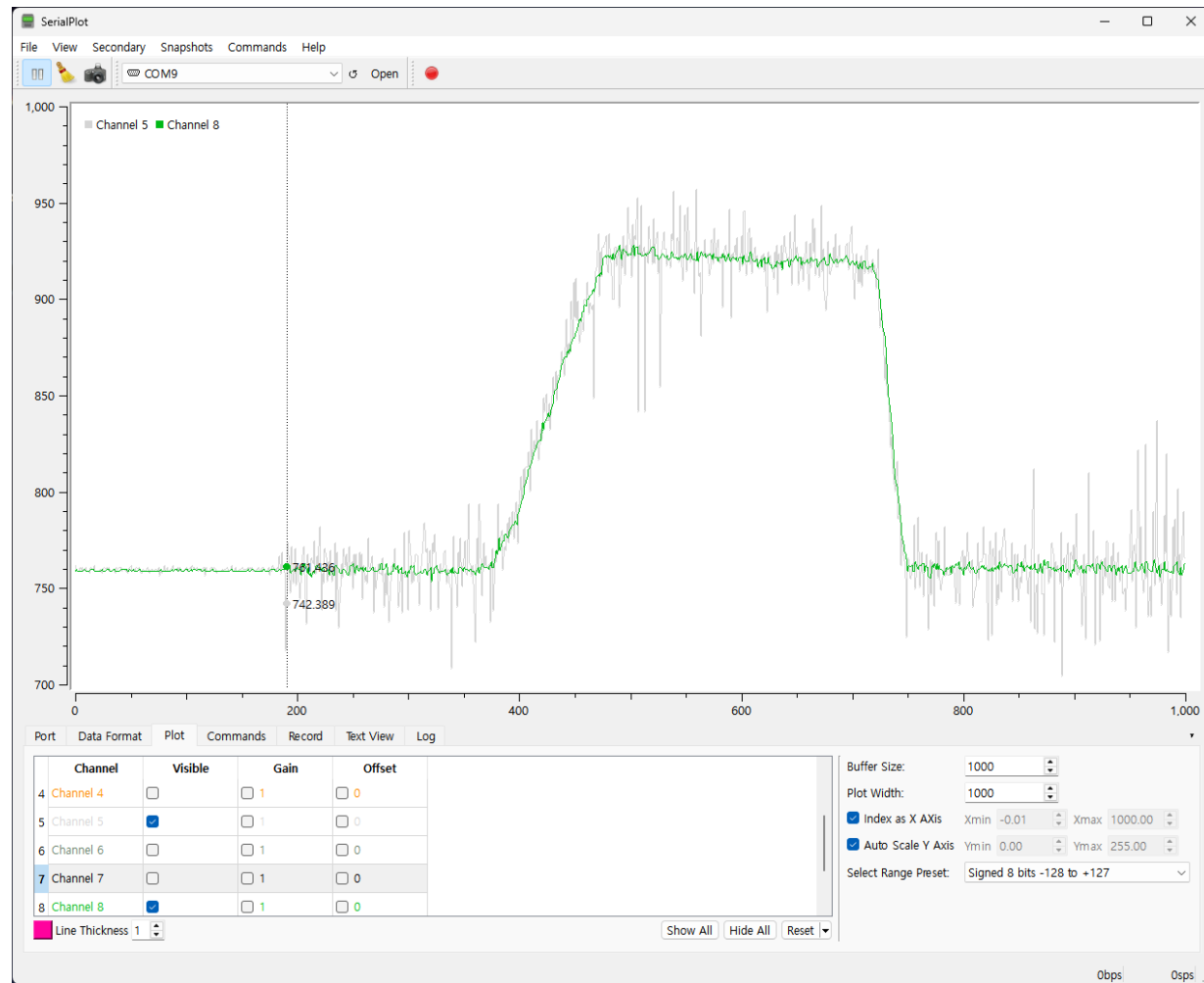
EMI 유입으로 인한 노이즈 현상



모달리티 선정과 어플리케이션에 대한 고민



Crawling robot – 단일 센서로 모든 필요한 모달리티 자가 인지 + 셀프 충전
Weakly electric fish
터치리스 인터페이스/로봇 안전센서



PINN(Physics Informed Neural Networks)

연구 중인 AI-LMS(액체금속 센서)의 디커플링 문제에 PINN을 적용하기 위한 아키텍처와 손실 함수(Loss Function) 설계입니다.

[Step 1] 신경망의 정의 (Forward Pass)

센서에서 측정된 저항 변화율(ΔR)과 인덕턴스 변화율(ΔL)을 입력받아 변형률($\hat{\epsilon}$)과 거리($\hat{d}$)를 예측하는 가중치 θ 로 구성된 신경망 N_θ 를 정의합니다.

$$[\hat{\epsilon}, \hat{d}] = N_\theta(\Delta R, \Delta L)$$

[Step 2] 데이터 손실 함수 (Data Loss)

실험으로 확보한 Ground Truth 데이터(ϵ, d)와 예측값의 차이를 계산합니다.

$$Loss_{Data} = \frac{1}{N} \sum_{i=1}^N \left(\|\hat{\epsilon}_i - \epsilon_i\|^2 + \|\hat{d}_i - d_i\|^2 \right)$$

[Step 3] 물리 손실 함수 (Physics Loss) 구축

신경망이 예측한 $\hat{\epsilon}$ 과 $\hat{d}$ 를 제안하신 물리적 지배 방정식의 역방향으로 다시 대입하여, 이론적인 저항(R_{theory})과 인덕턴스(L_{theory})를 계산합니다.

- 저항 이론값:

$$R_{theory}(\hat{\epsilon}) = \alpha_1 \hat{\epsilon} + \alpha_2 \hat{\epsilon}^2$$

- 인덕턴스 이론값:

$$L_{theory}(\hat{\epsilon}, \hat{d}) = \beta_1 \hat{\epsilon} + \frac{\beta_2}{(\hat{d} + d_0)^k} + \beta_3 \hat{\epsilon} \cdot f(\hat{d})$$

이 이론값이 실제 센서에 입력으로 들어왔던 $\Delta R, \Delta L$ 과 일치해야 물리 법칙이 성립하므로, 이 차이를 물리적 페널티로 줍니다.

$$Loss_{Physics} = \frac{1}{N} \sum_{i=1}^N \left(\|\Delta R_i - R_{theory}(\hat{\epsilon}_i)\|^2 + \|\Delta L_i - L_{theory}(\hat{\epsilon}_i, \hat{d}_i)\|^2 \right)$$

[Step 4] 최종 목적 함수 (Total Loss)

데이터 기반 오차와 물리 법칙 위반 오차를 합산하여 최종 가중치 θ 를 최적화합니다. λ 는 물리 손실의 가중치를 조절하는 하이퍼파라미터입니다.

$$Loss_{Total} = Loss_{Data} + \lambda \cdot Loss_{Physics}$$

이 수식을 통해 모델은 데드존(d 가 커서 ΔL 의 변화가 거의 없는 구간)에 진입하더라도, 물리 수식의 뼈대(L_{theory})를 따라가려는 성질 때문에 발산하지 않고 가장 물리적으로 타당한 거리를 스스로 추론해 냅니다.

푸바오가 돌아왔다

